MANCHESTER 1824
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EART11100

How to Build a Planet

Week 2

The Heavy Construction: The Formation of the Stars and Galaxies

13650-4567 Ma

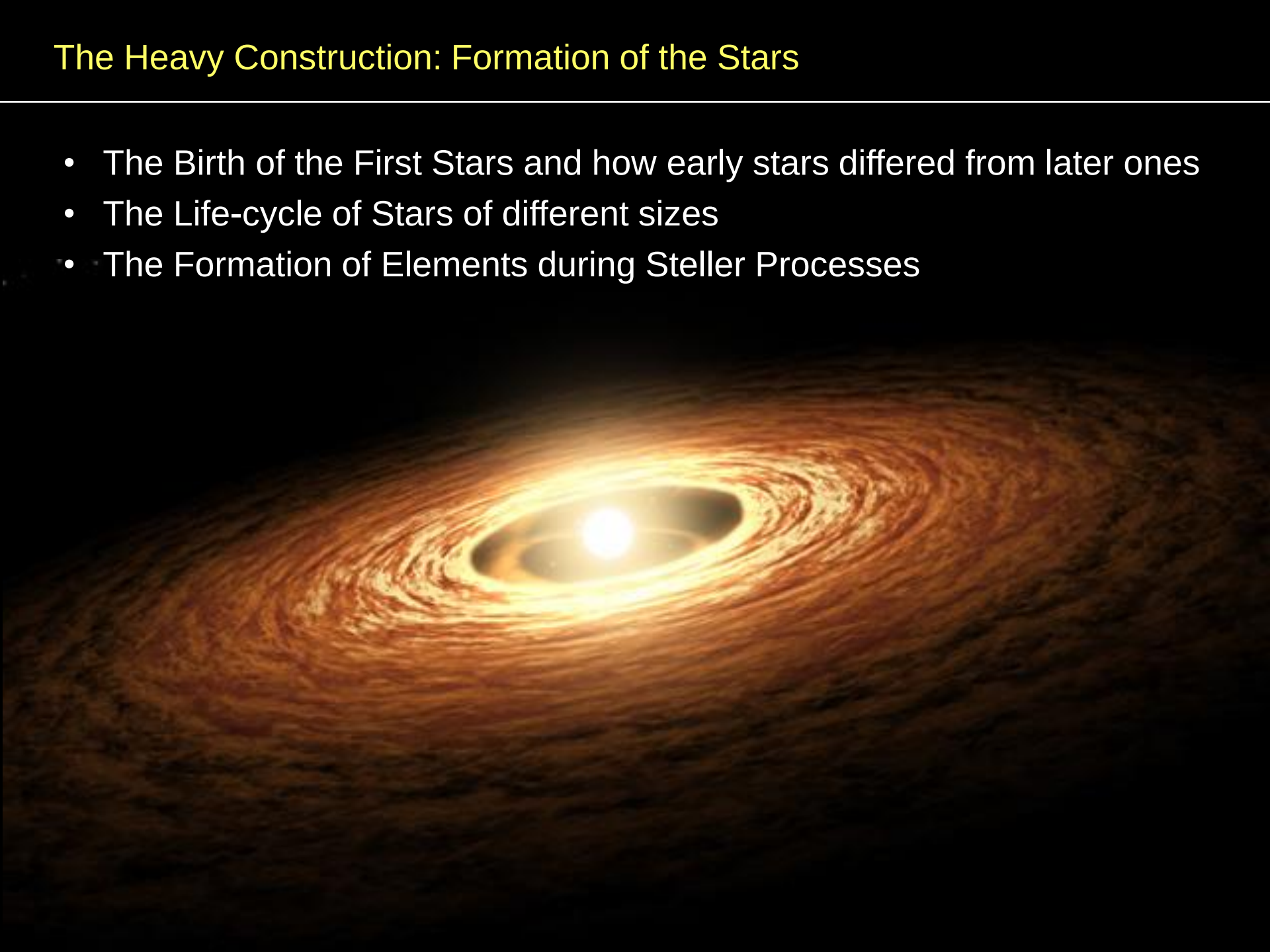
Outline

Part 1 : Formation and lifecycle of stars

Part 2 : The Milky Way Galaxy

The Heavy Construction: Formation of the Stars

- The Birth of the First Stars and how early stars differed from later ones
- The Life-cycle of Stars of different sizes
- The Formation of Elements during Steller Processes



Recap: The three pillars of Big Bang theory

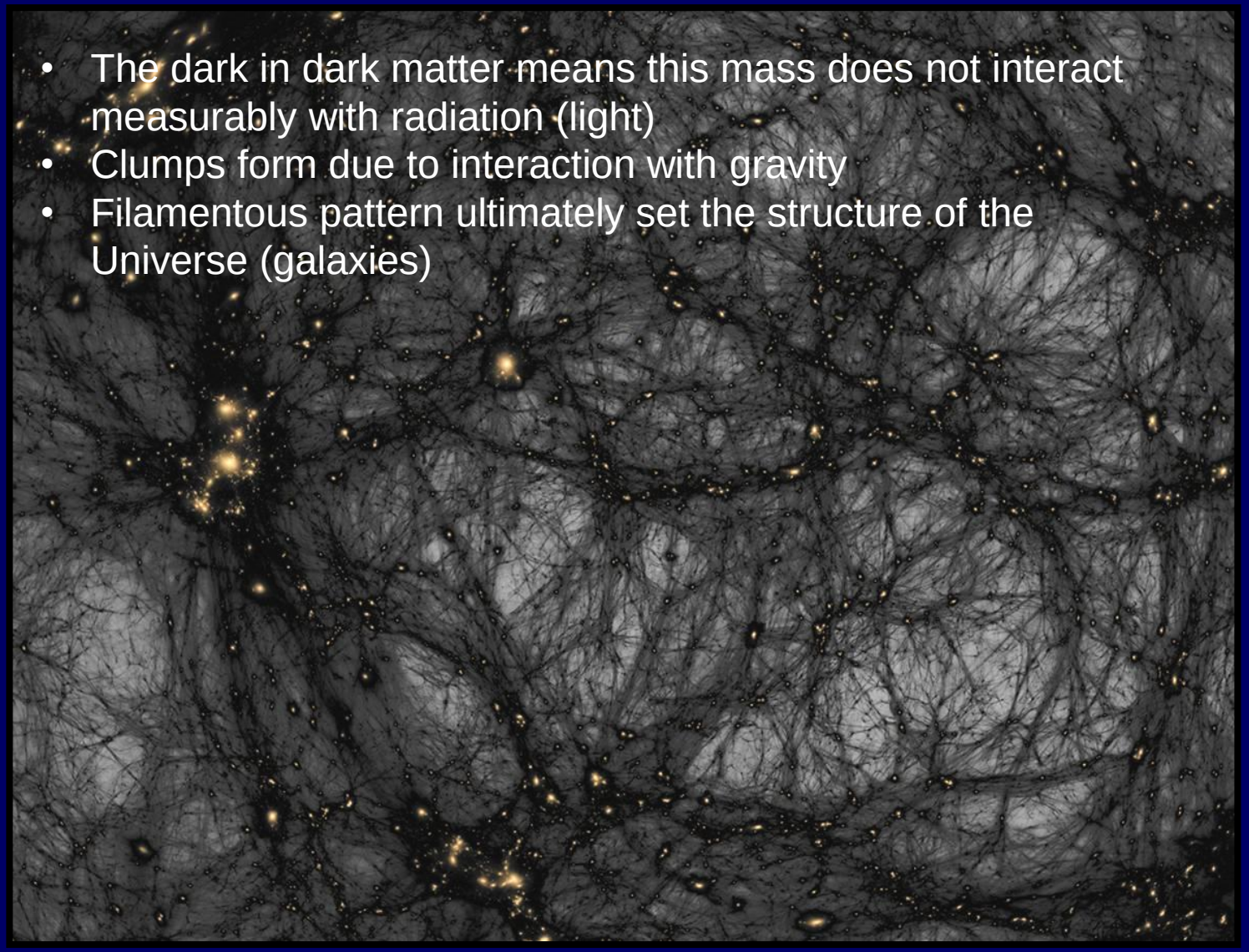
1. The observed expansion of the Universe (cosmological redshift).

2. That hydrogen and helium are very abundant in the Universe, since they - unlike other elements - formed during Big Bang Nucleosynthesis (BBN).

3. The properties of the Cosmic Microwave Background (recombination of hydrogen and escape of photons).

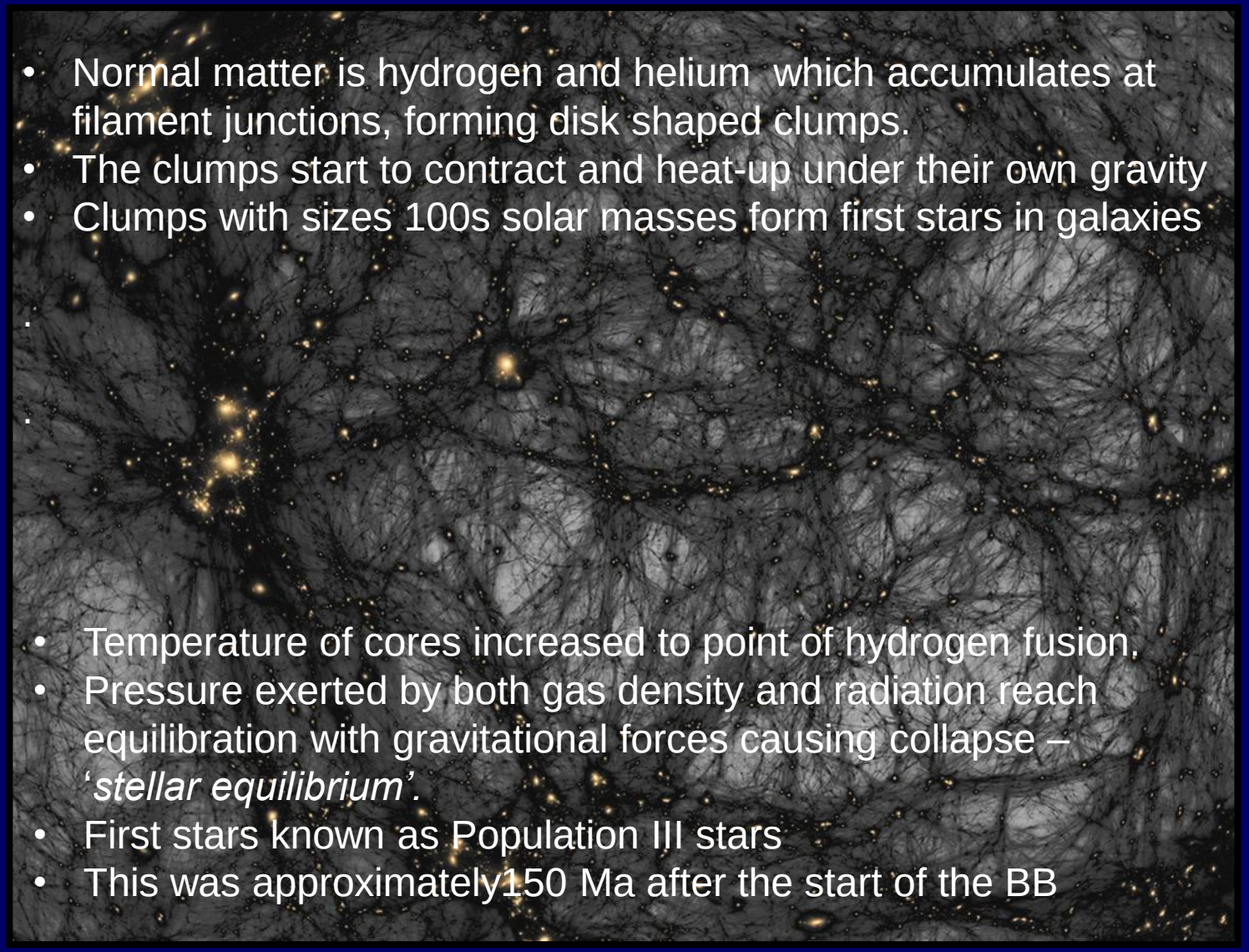
The Big Bang occurred around 13.7 billion years ago.

Cosmic Dark Ages: 380 kyr < *t* < 150 myr



A simulation of the distribution of dark matter (dark areas) across a region of Universe

Cosmic Dark Ages: 380 kyr < *t* < 150 myr



- Normal matter is hydrogen and helium which accumulates at filament junctions, forming disk shaped clumps.
- The clumps start to contract and heat-up under their own gravity
- Clumps with sizes 100s solar masses form first stars in galaxies

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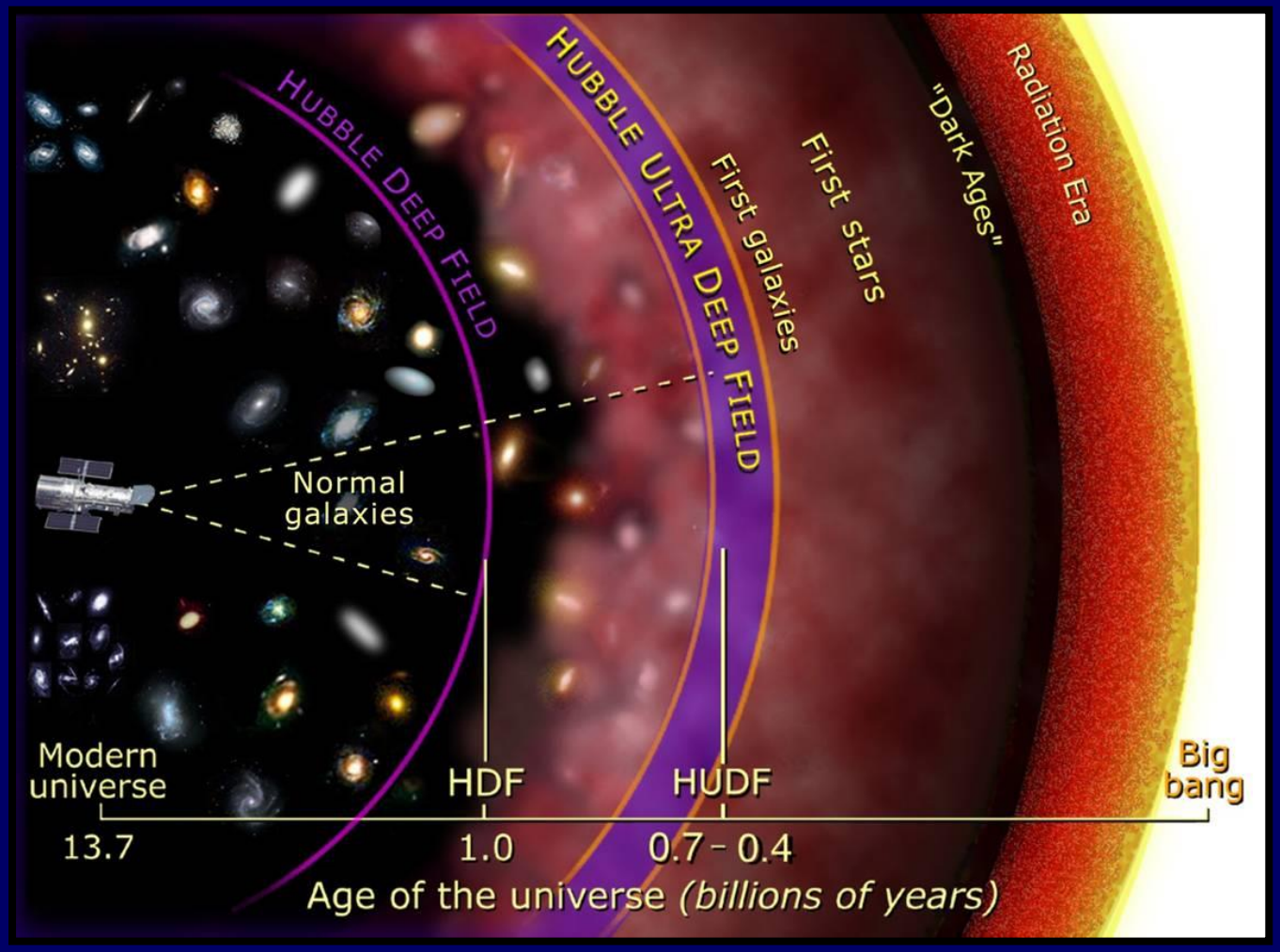
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- Temperature of cores increased to point of hydrogen fusion.
- Pressure exerted by both gas density and radiation reach equilibration with gravitational forces causing collapse – '*stellar equilibrium*'.
- First stars known as Population III stars
- This was approximately150 Ma after the start of the BB

A simulation of the distribution of dark matter (dark areas) across a region of Universe

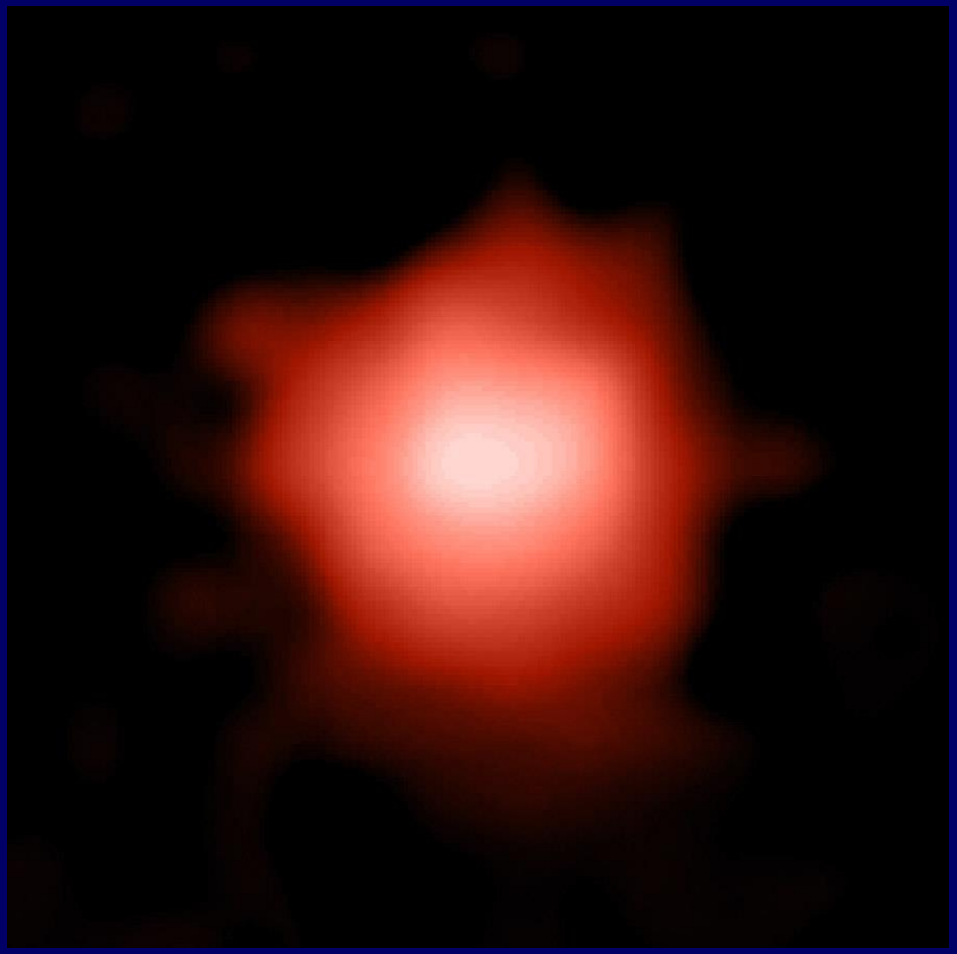
The first stars (Population III) t = 150 myr

- The first stars were short-lived (few million years), supermassive (100+ solar masses), Population III stars → within the star interiors, the first heavy elements were formed and these were expelled into space in giant supernovas once fuel supplies were exhausted



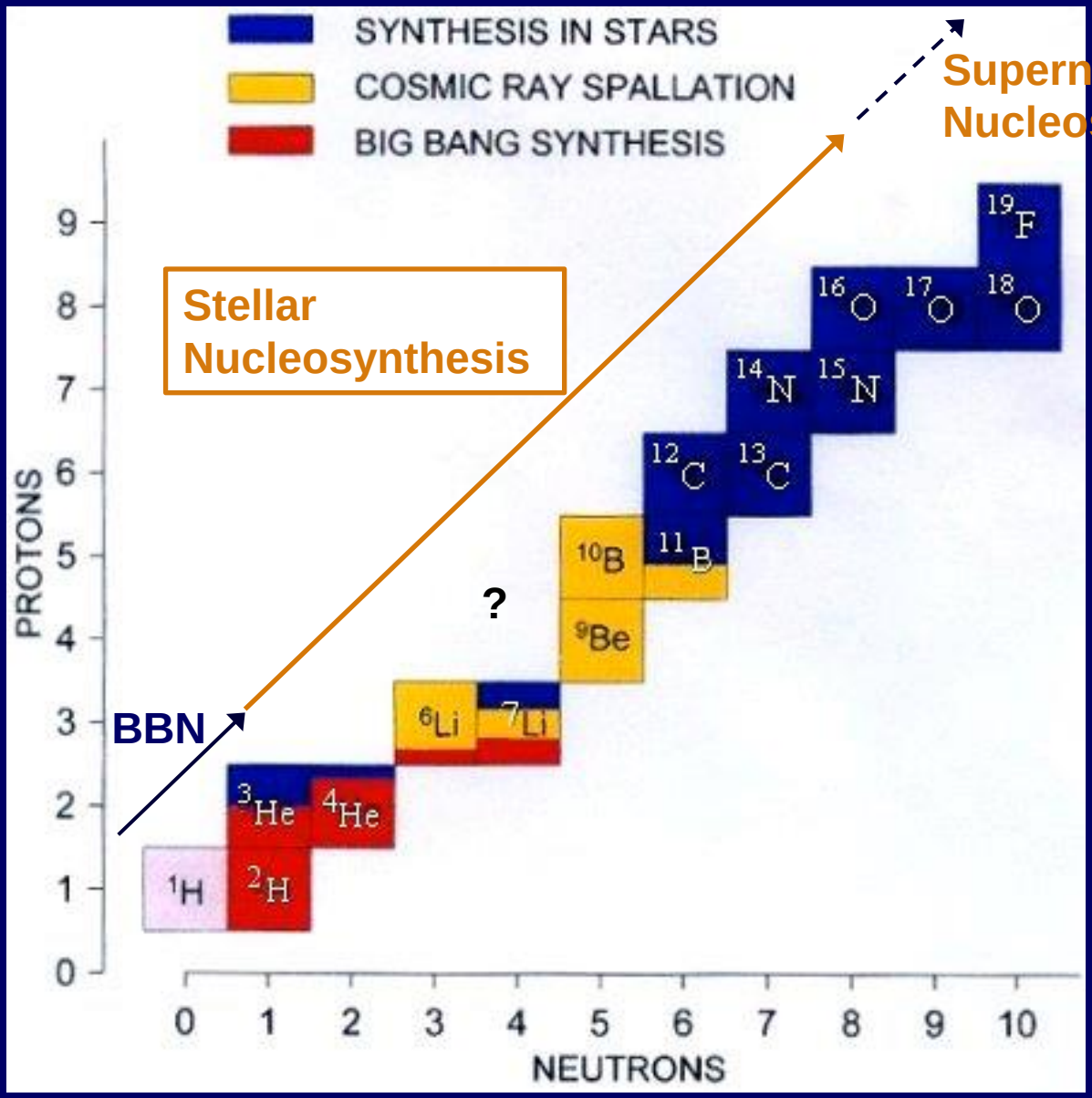
The deepest field images from the Hubble telescope do not extend quite far enough back to show the earliest stars; the James Webb Telescope will!

James Webb Space Telescope (JWST)



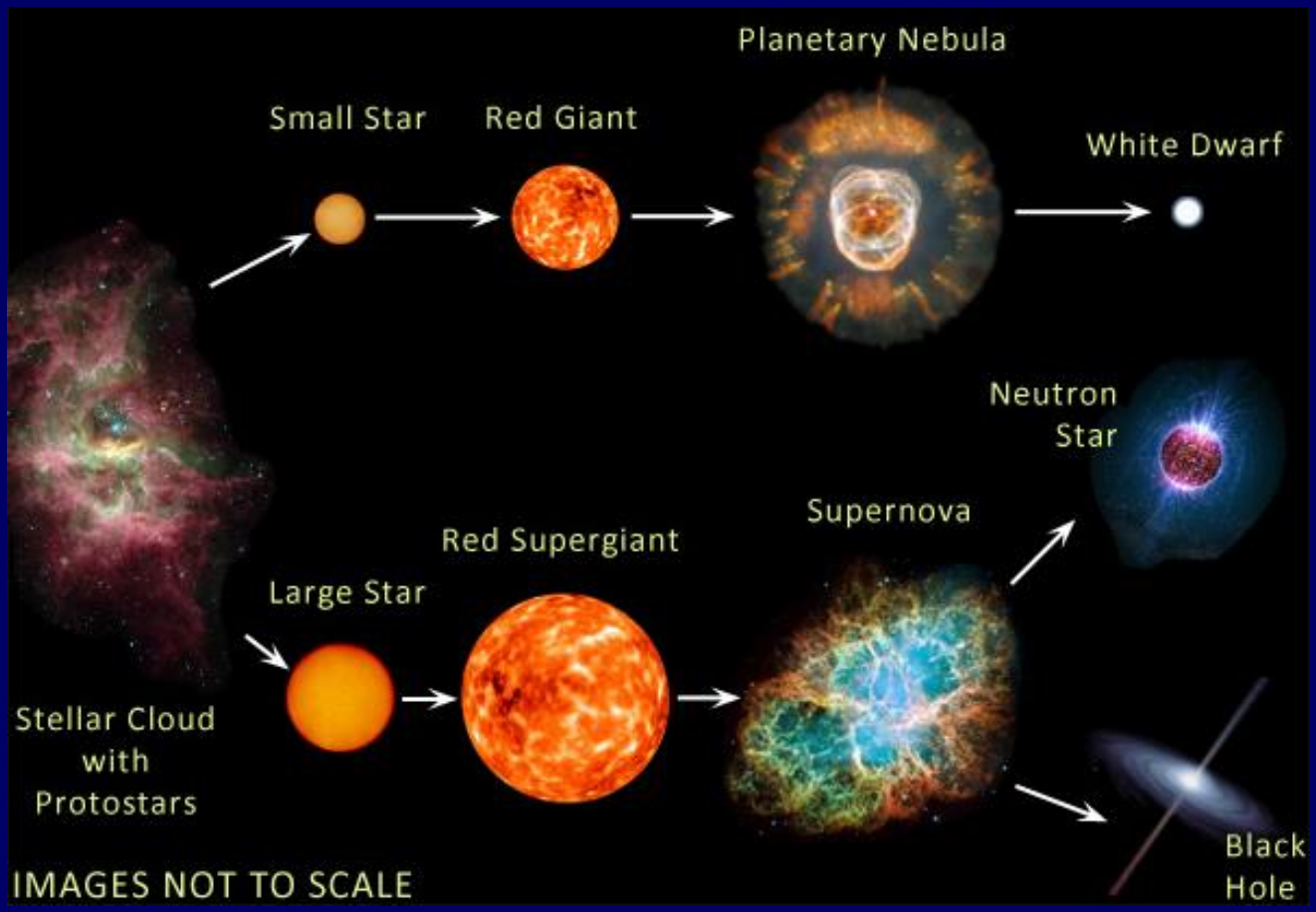
Newly discovered galaxy GLASS-z13, which is so far away that we see it as it appeared 300 million years after the Big Bang, now holds the record for the earliest known galaxy. (Image NASA/CSA/ESA/STScI)

Stellar nucleosynthesis



The life-cycle of a star depends on its mass

Through the life cycle of a star, there is a balance between expansion promoted by the fusion reactions and contraction due to gravity



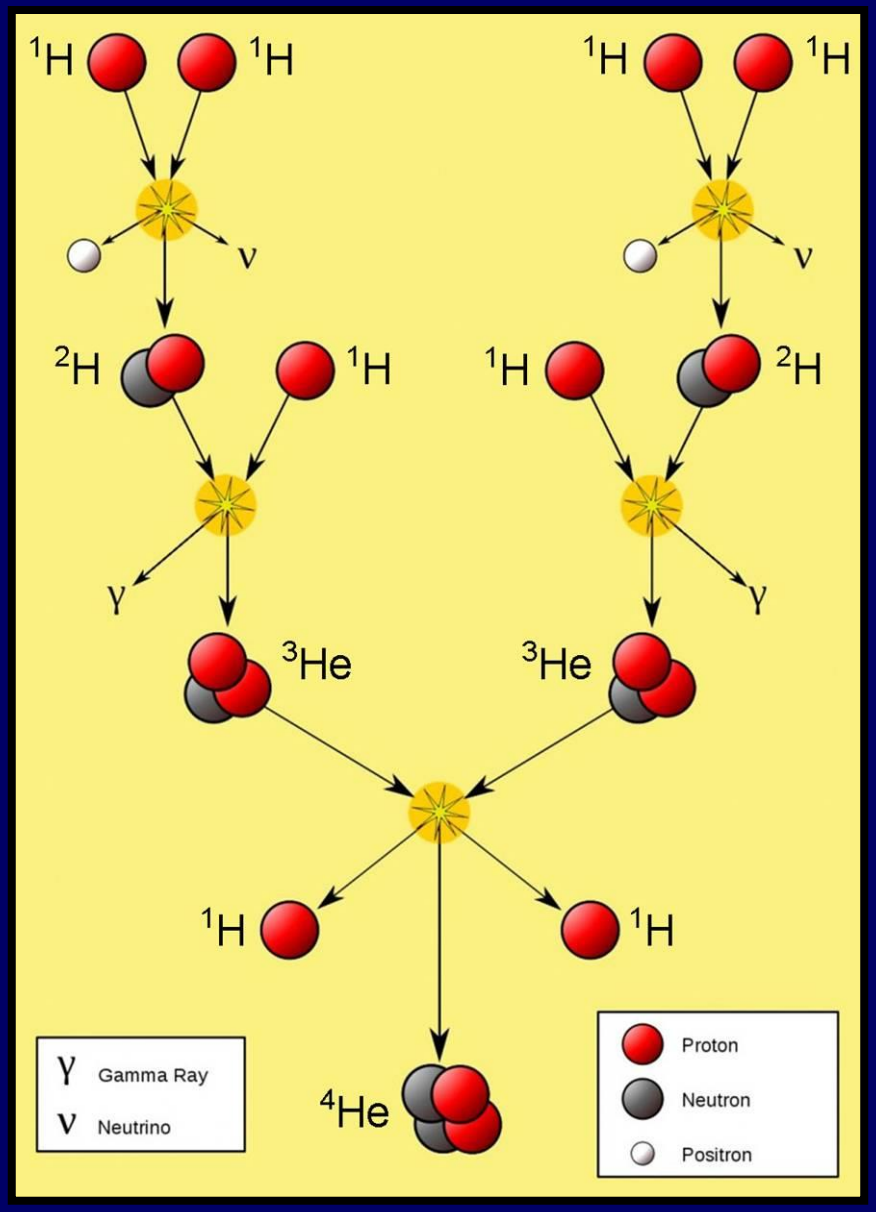
Sun-sized stars

Massive stars

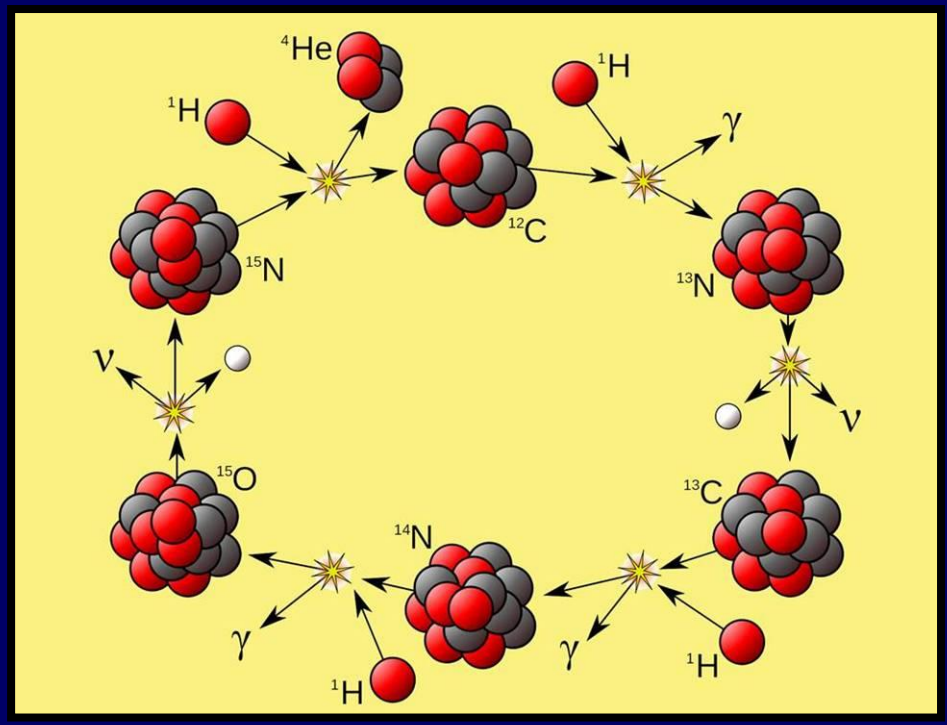
Contraction → increase core T → promoting fusion reactions → expansion

Expansion → decrease core T → inhibiting fusion reactions → gravity collapse

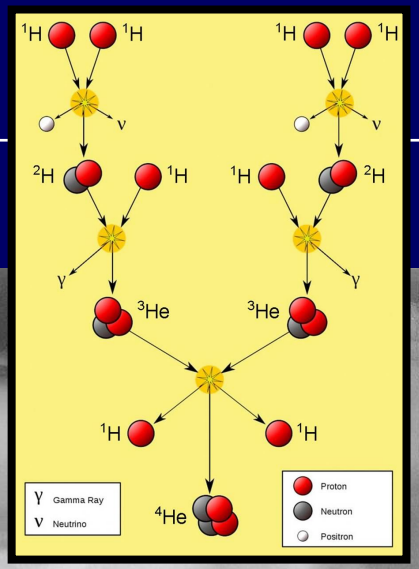
Hydrogen burning process



- Fusion through most of the life of a star is by a proton-proton chain reaction that uses four protons to produce one helium-4 nucleus
- Under hotter conditions (more massive stars), hydrogen burning to form helium is more efficiently done using carbon-12 as a catalyst (CNO cycle)



The life cycle of Sun-sized stars: nuclear fusion

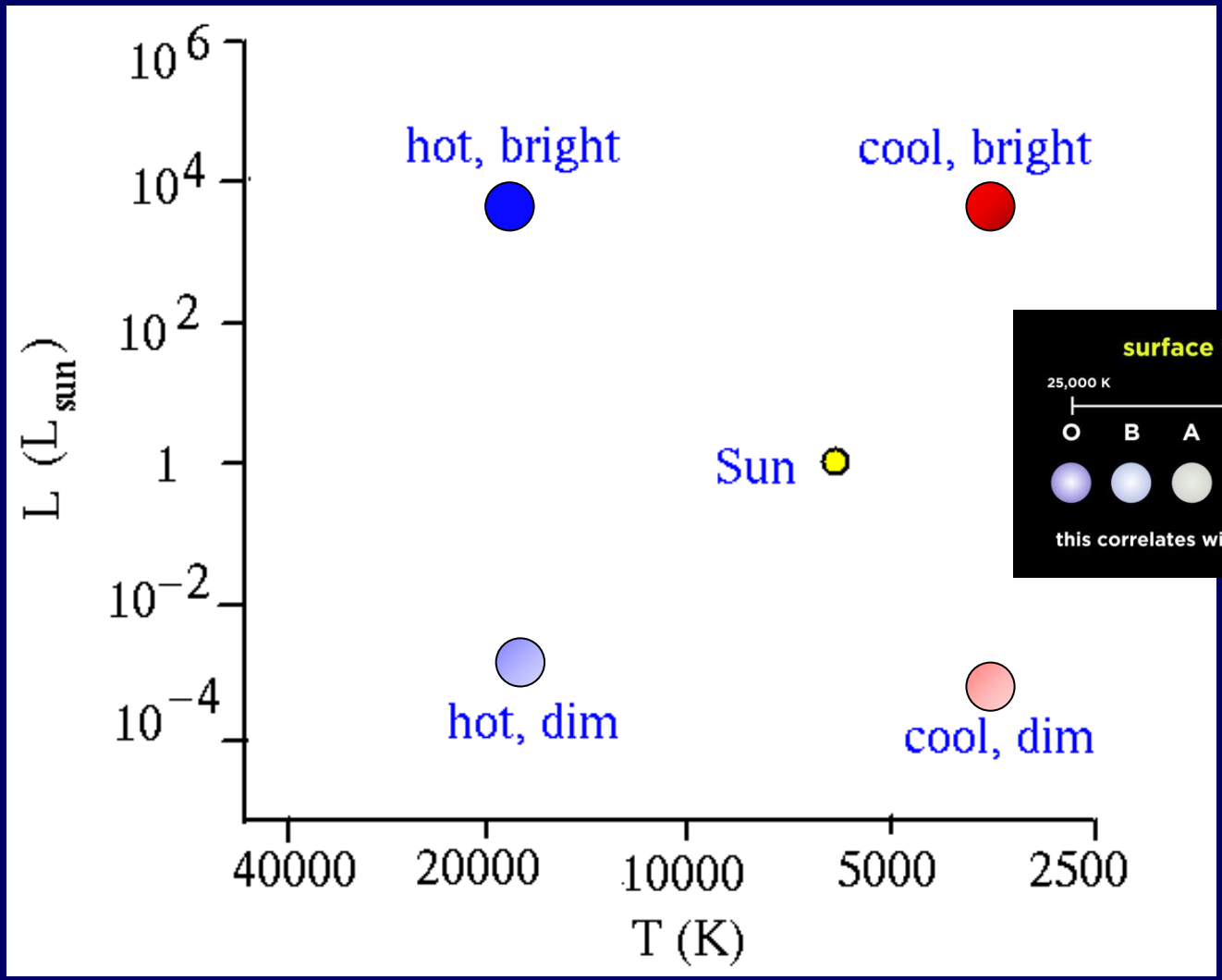


- The mass of an atom is always slightly less than the sum of the masses of its individual neutrons, protons and electrons = mass defect, ΔM.
- Some mass is used to bind the nucleus = binding energy, E_b
- During a fusion reaction the mass defect is converted to energy

$E_b = \Delta Mc^2$ c = speed of light (2.997925×10^{10} cm/s)

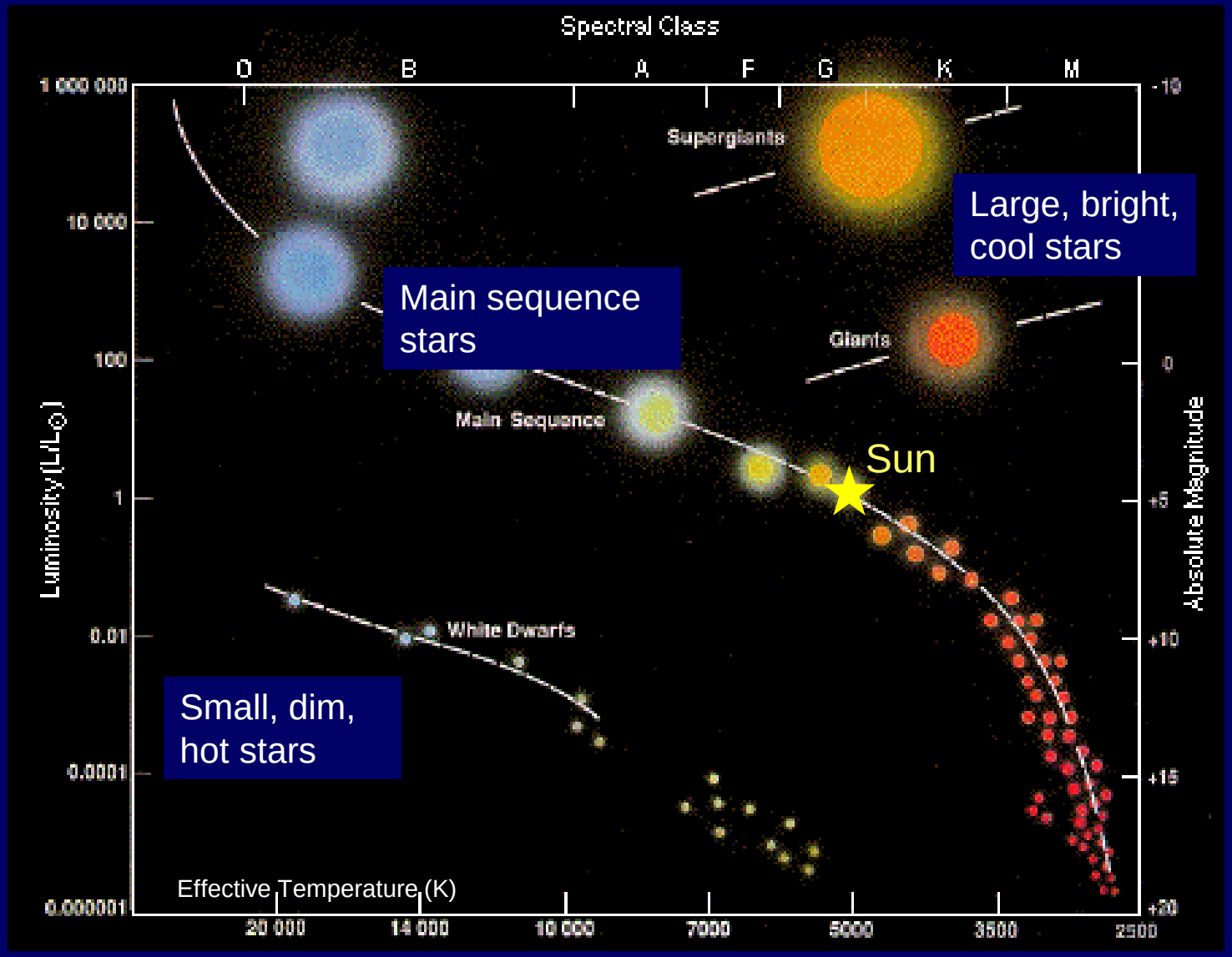
Transformation of Protostar to Main-sequence star

The life cycle of Sun-sized stars



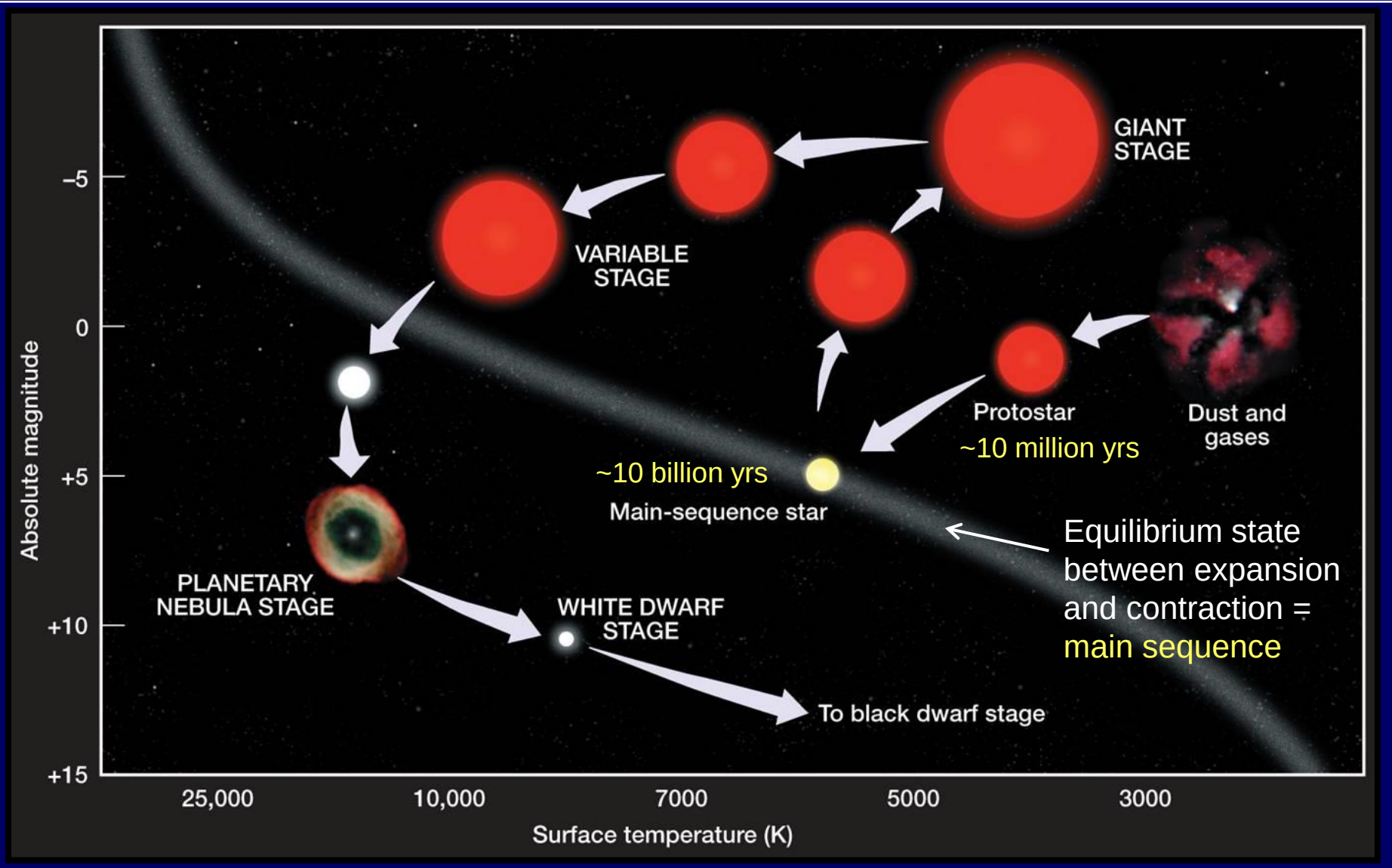
Variation in star spectrum & surface temperature on a Hertzsprung-Russell (H-R) diagram

The life cycle of Sun-sized stars (HR diagram)



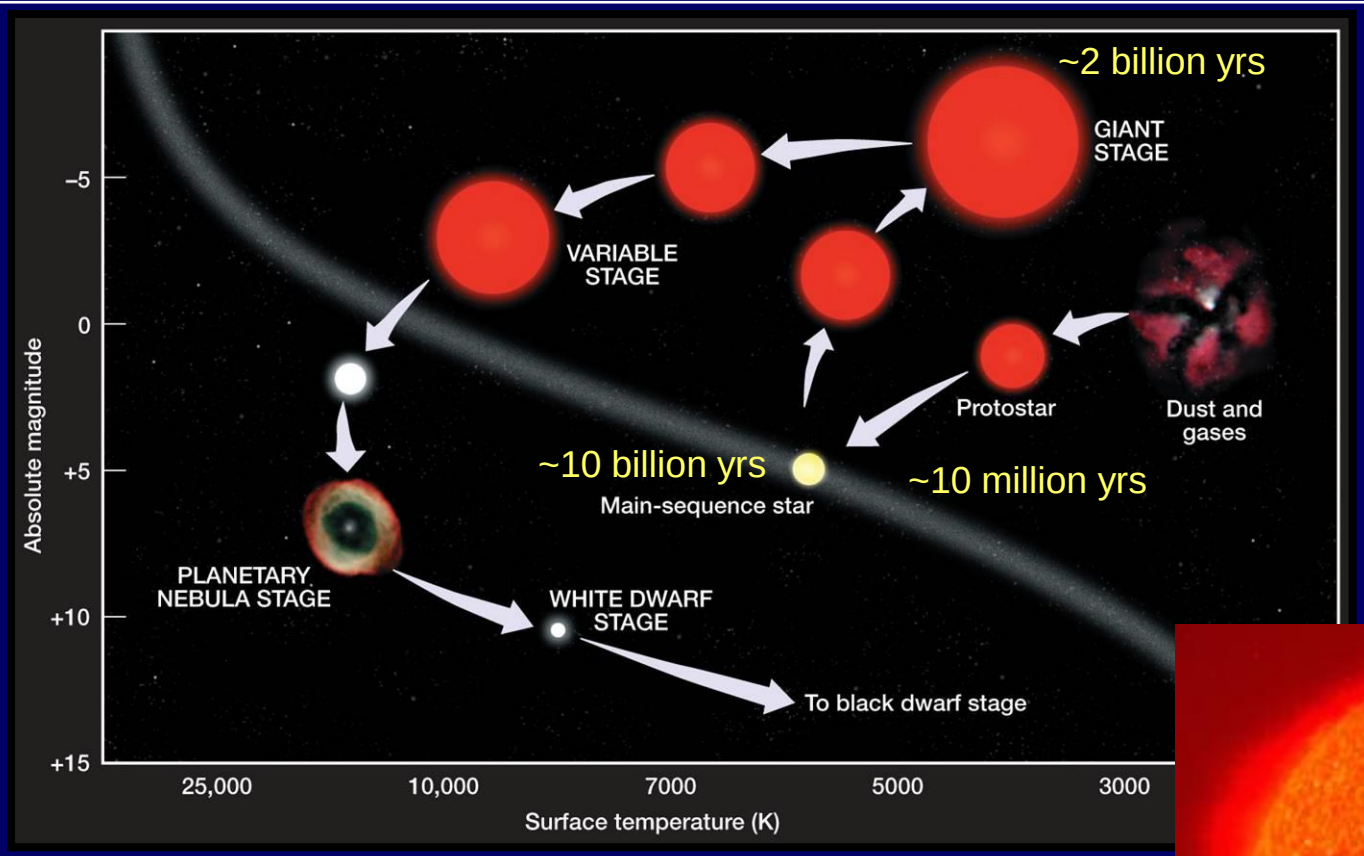
Variation in star colour (spectrum) and size on a Hertzsprung-Russell diagram

The life cycle of Sun-sized stars (HR diagram)



Our sun: main sequence star, ~ ½ through life

The life cycle of Sun-sized stars



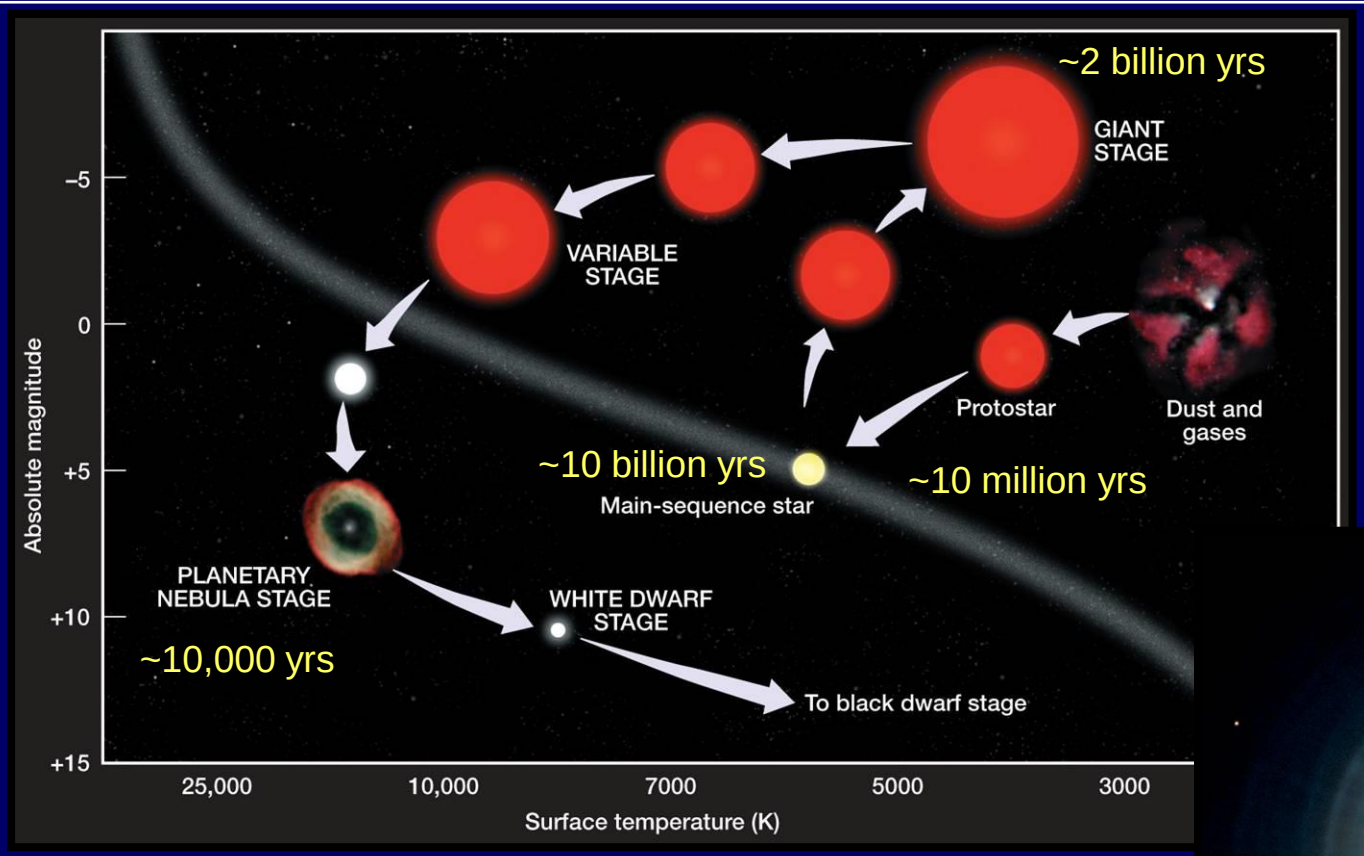
Track of a star on a Hertzsprung-Russell diagram

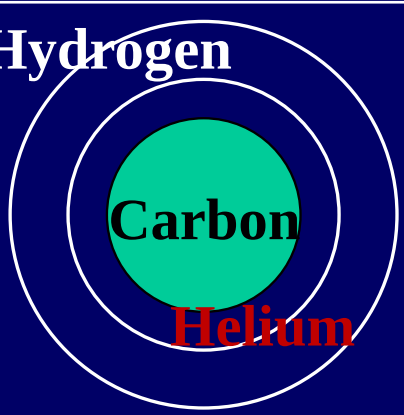


Transformation to Red Giant:

- core collapse, raising T → H burning in shell → star swells (red giant) and T increases → He burning in core to form C

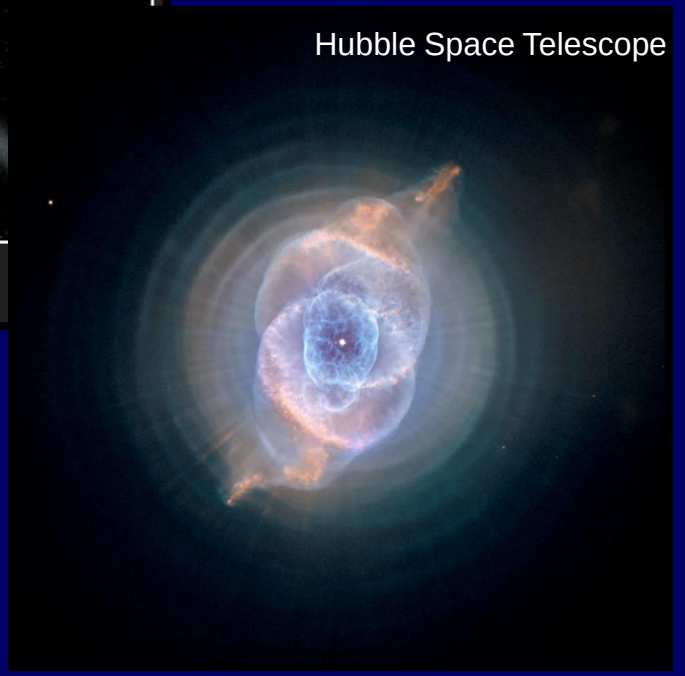
The life cycle of Sun-sized stars





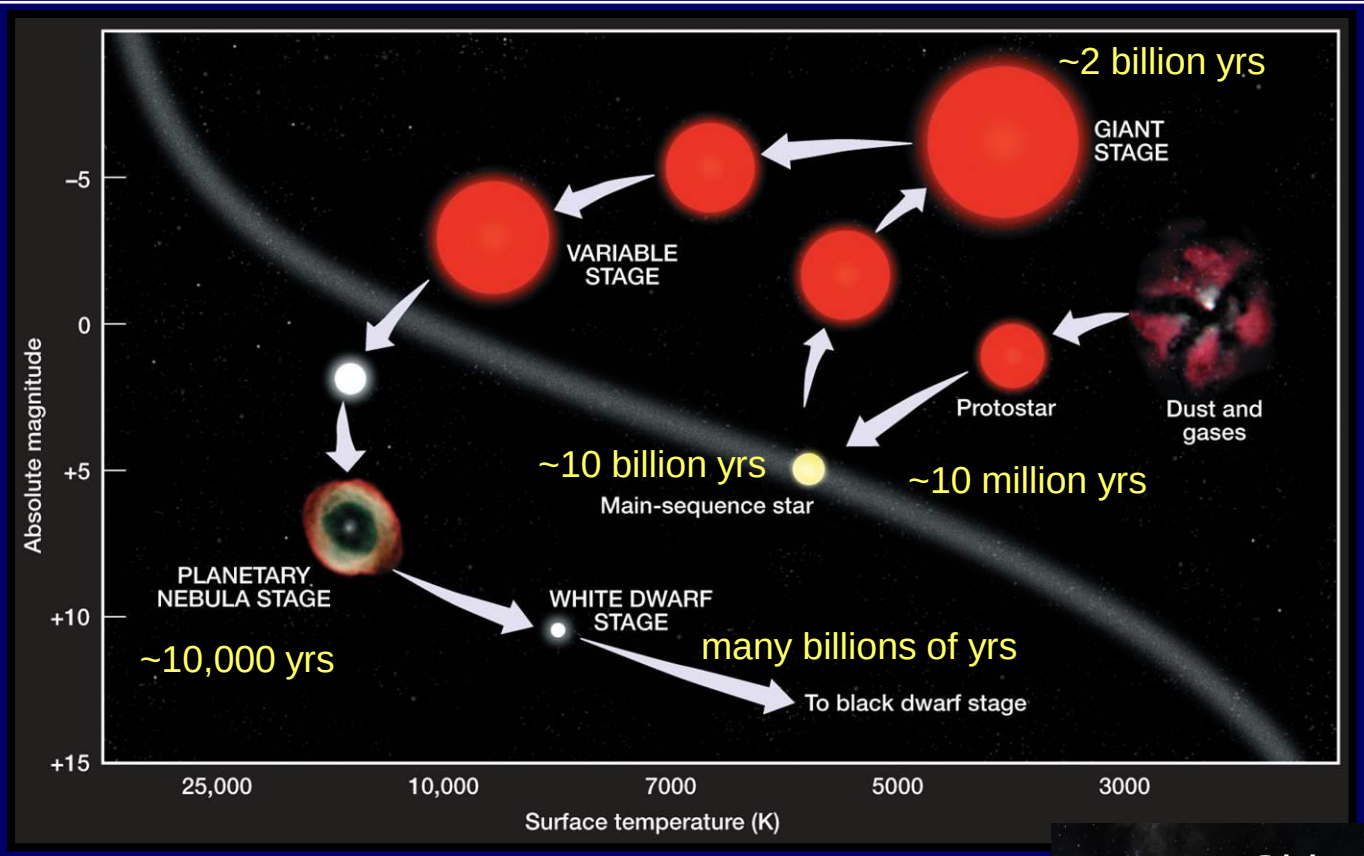
Transformation to Variable star and then Planetary Nebula:

- when all He in core has been converted to C, He shell burning
- → as H and He decline, pulsed burning (variable stage)
- shedding outer layers into space (planetary nebula stage)



Planetary nebula (the Cat's Eye nebula)

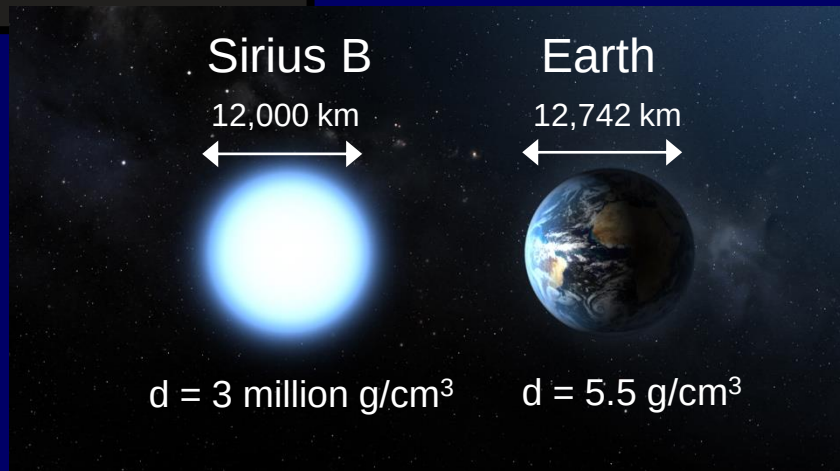
The life cycle of Sun-sized stars



Track of a star on a Hertzsprung-Russell diagram

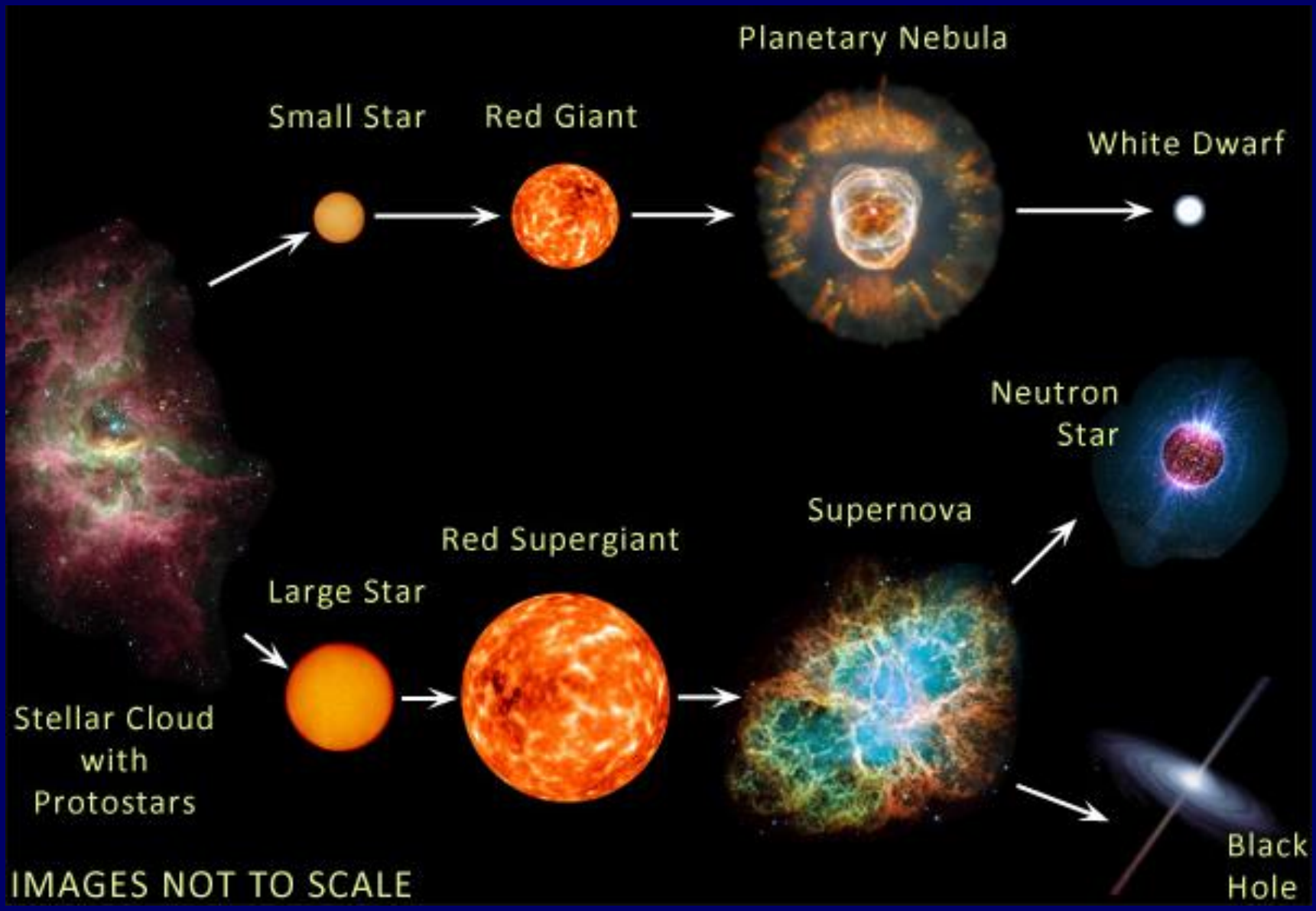
Transformation to White Dwarf and then Black Dwarf star:

- radiative cooling to a fading husk (dwarf stage)
- collapse ends when ~100 times smaller than main-sequence star
- electron degeneracy pressure



White dwarf (Sirius B)

The end stage of massive stars is different

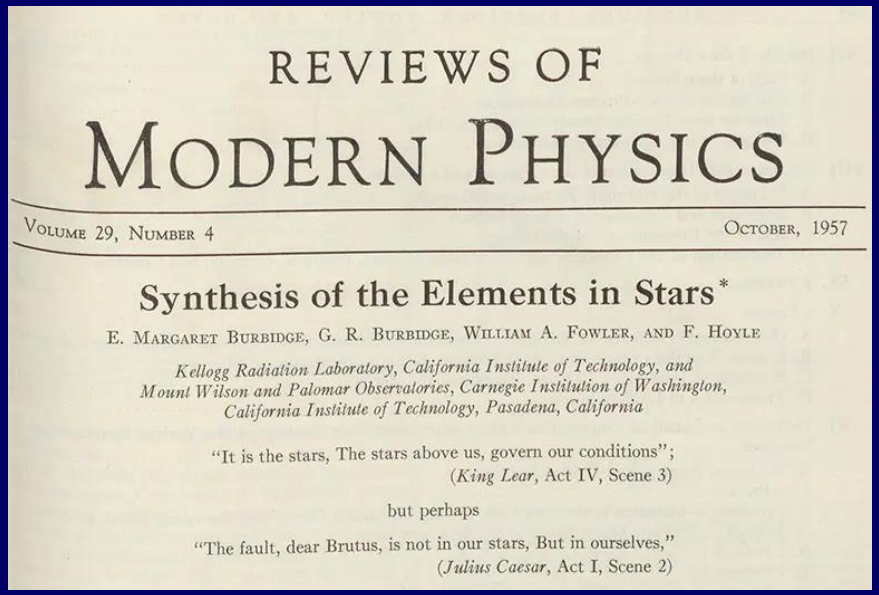


Fred Hoyle

- "Synthesis of the Elements in Stars" published in 1954, better known as B^2FH from the initials of its authors: Margaret Burbidge, Geoffrey Burbidge, William A. Fowler, and Fred Hoyle.



Image : https://www.joh.cam.ac.uk/library/special_collections/hoyle/exhibition/b2fh



REVIEWS OF

MODERN PHYSICS

VOLUME 29, NUMBER 4 OCTOBER, 1957

Synthesis of the Elements in Stars*

E. MARGARET BURBIDGE, G. R. BURBIDGE, WILLIAM A. FOWLER, AND F. HOYLE

Kellogg Radiation Laboratory, California Institute of Technology, and Mount Wilson and Palomar Observatories, Carnegie Institution of Washington, California Institute of Technology, Pasadena, California

"It is the stars, The stars above us, govern our conditions";
(*King Lear*, Act IV, Scene 3)

but perhaps

"The fault, dear Brutus, is not in our stars, But in ourselves,"
(*Julius Caesar*, Act I, Scene 2)

Stellar nucleosynthesis (fusion in massive stars)

Temperatures required to initiate the various fusion reactions:

- **hydrogen burning** by the proton-proton chain reaction, ~4 x 10^6 K
- **hydrogen burning** by the CNO cycle, 1.6 x 10^7 K
- **helium burning** by the triple alpha process (producing ^{12}C and some ^{16}O), 2.3 x 10^8 K

Stars over about 8-solar masses

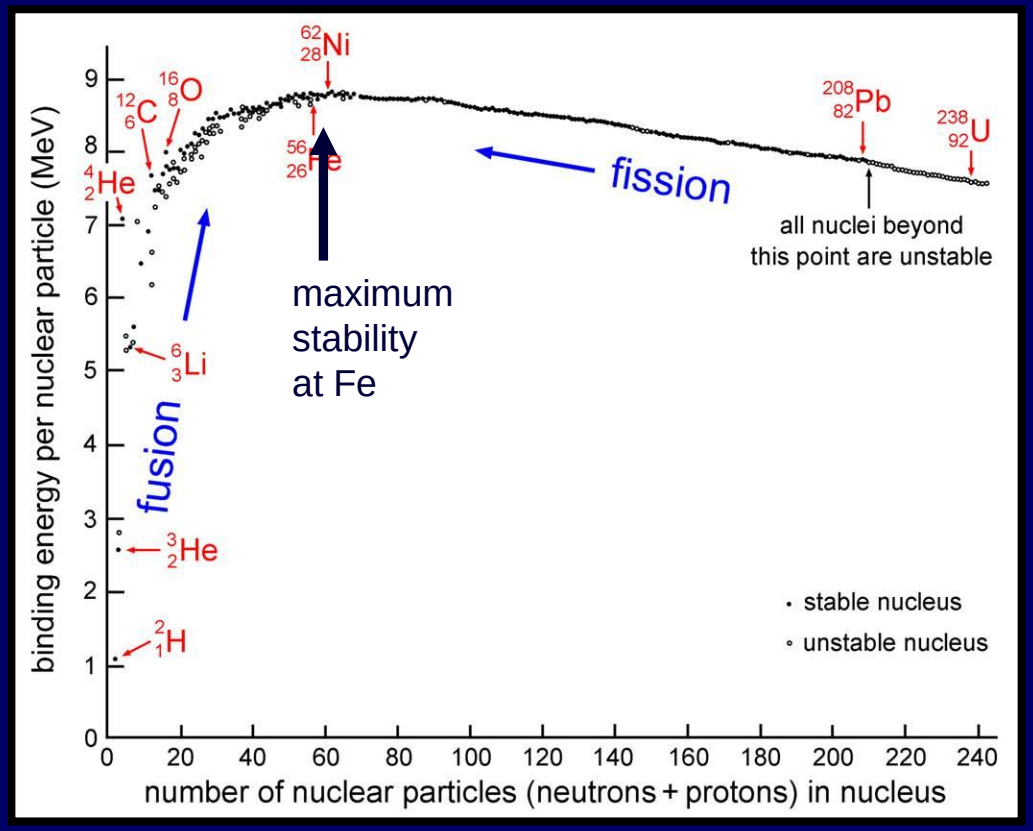
- **carbon burning** (producing ^{20}Ne, ^{23}Na, ^{23}Mg, ^{24}Mg, and ^{16}O), 9.3 x 10^8 K
- **neon burning** (producing ^{16}O and ^{24}Mg), 1.7 x 10^9 K

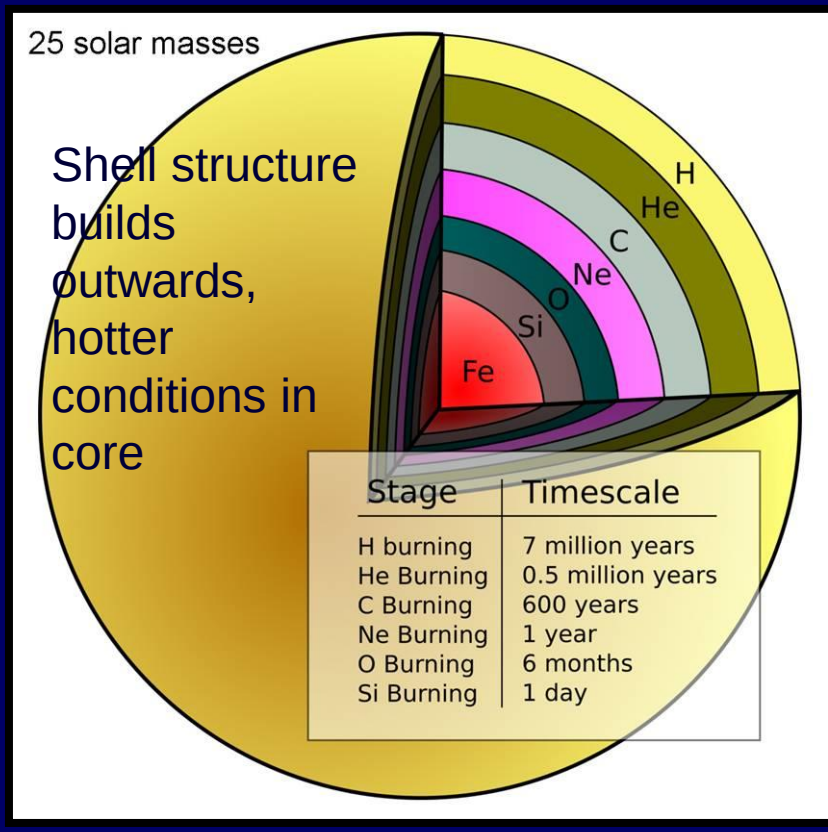
Stars over about 10-solar masses

- **oxygen burning** (producing ^{28}Si, ^{32}S, ^{24}Mg with some ^{31}P, ^{31}S, ^{30}Si, and ^{30}P), 2.3 x 10^9 K
- **silicon burning** by the alpha process (producing ^{32}S, ^{36}Ar, ^{40}Ca, ^{44}Ti, ^{48}Cr, ^{52}Fe, and $^{56}Ni \rightarrow {}^{56}Co \rightarrow {}^{56}Fe$), 4.1 x 10^9 K

Stellar nucleosynthesis (fusion in massive stars)

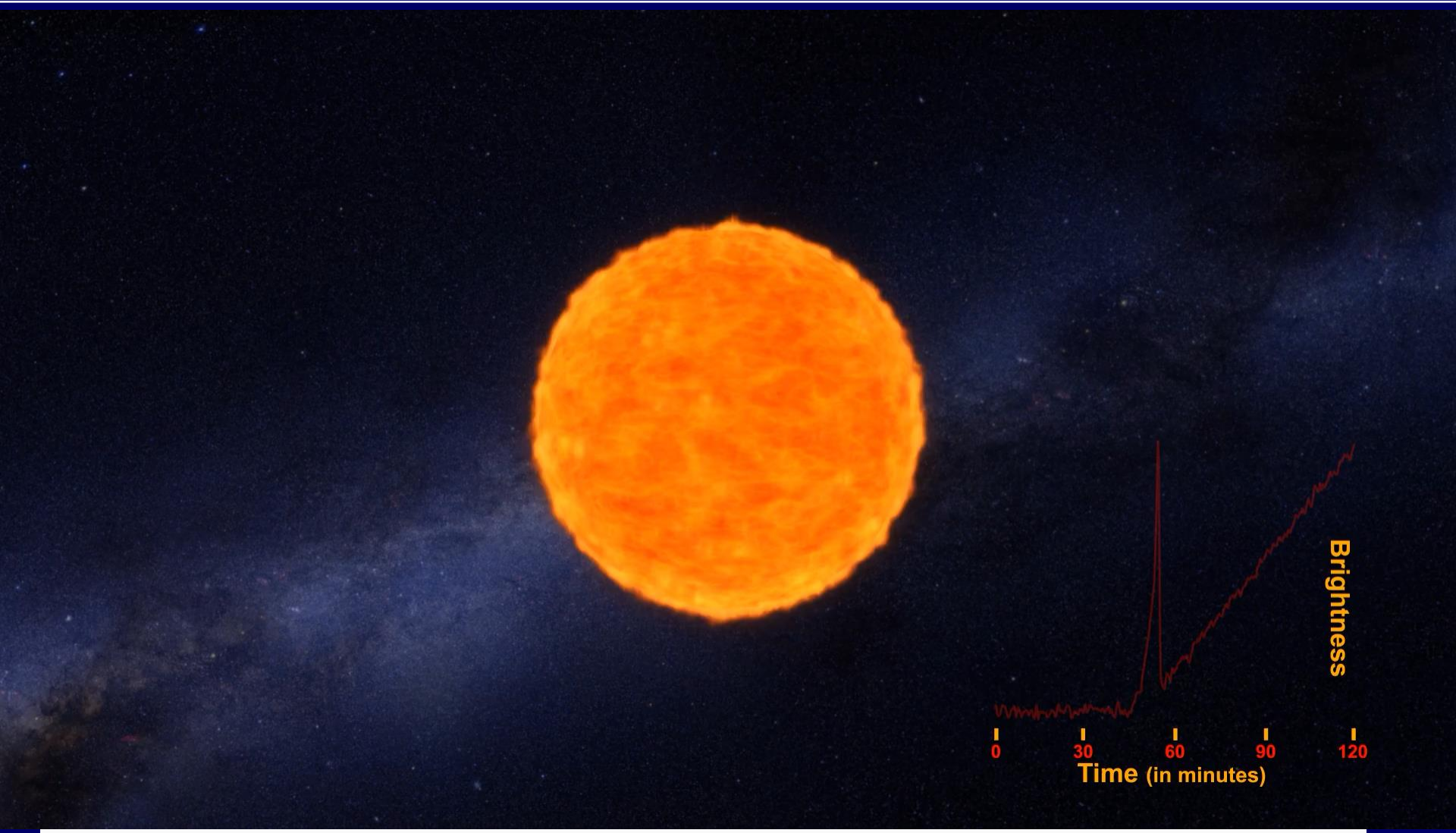
- Binding energy of nucleus is the driving force of fusion
- Fusion of elements lighter than iron may release energy to generate nuclei with greater binding energy (per nucleon).
- Fission of elements heavier than iron may release energy to generate nuclei with greater binding energy (per nucleon).





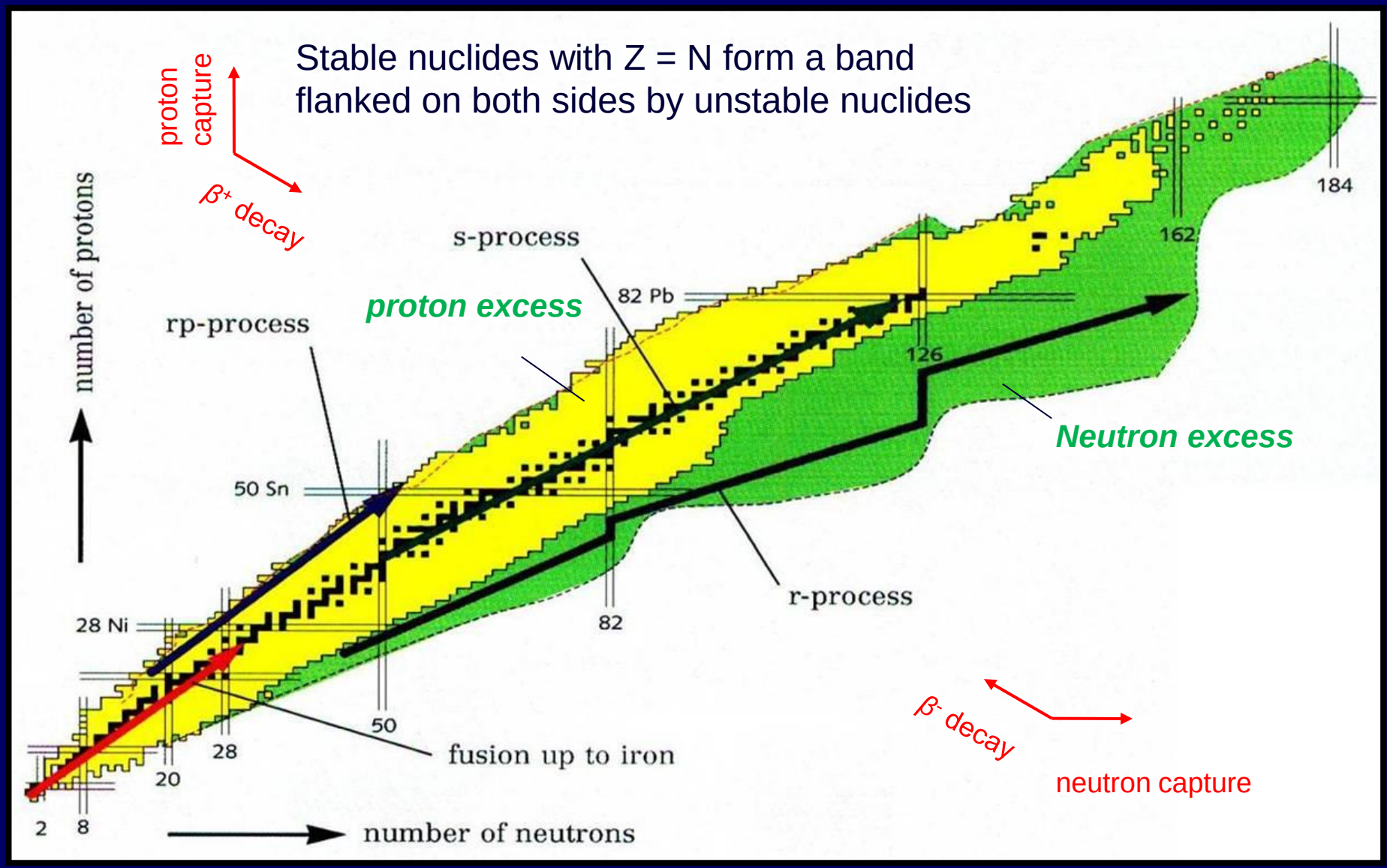
The core of a massive star in its last days

Explosion of a red supergiant



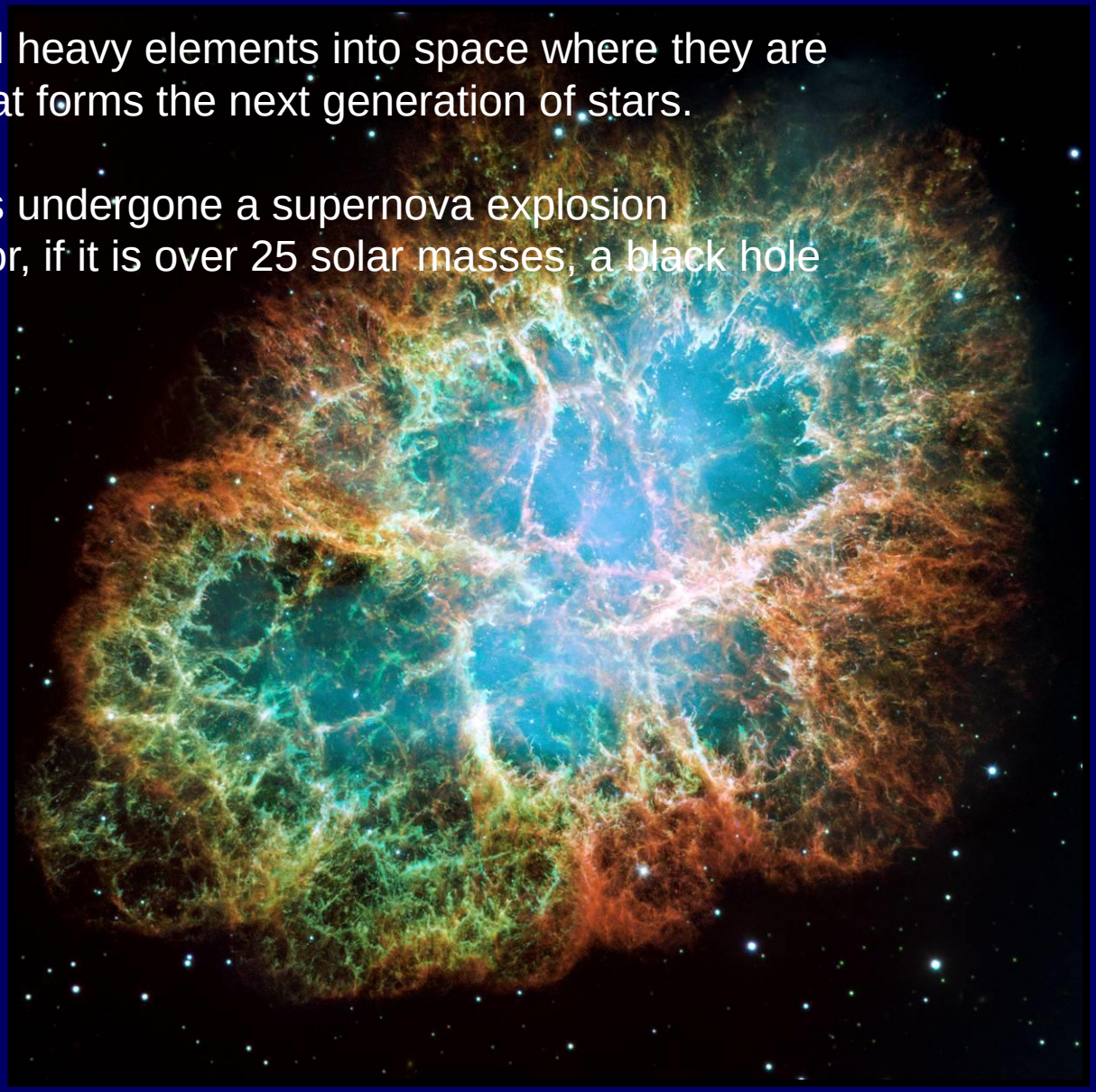
The early of an exploding red supergiant star (KSN 2011d), which is located 1.2 billion light years away, caught by Kepler in 2011 (Source: NASA Ames, STScI/G. Bacon)

Stellar nucleosynthesis (heavy elements)



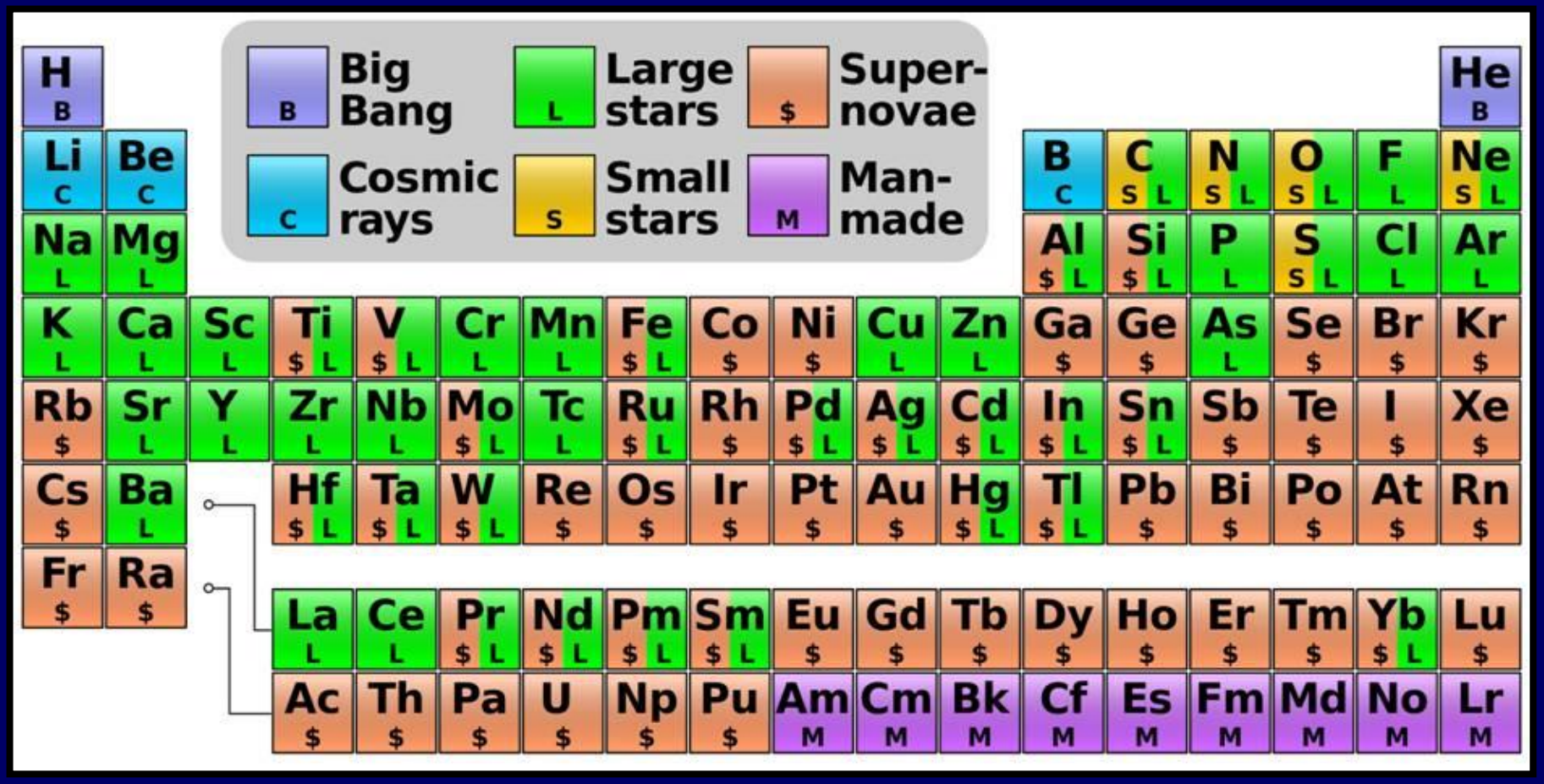
Stellar nucleosynthesis (supernova dispersal)

- Supernova explosions expel heavy elements into space where they are incorporated into the gas that forms the next generation of stars.
- The core of the star that has undergone a supernova explosion collapses to a neutron star or, if it is over 25 solar masses, a black hole



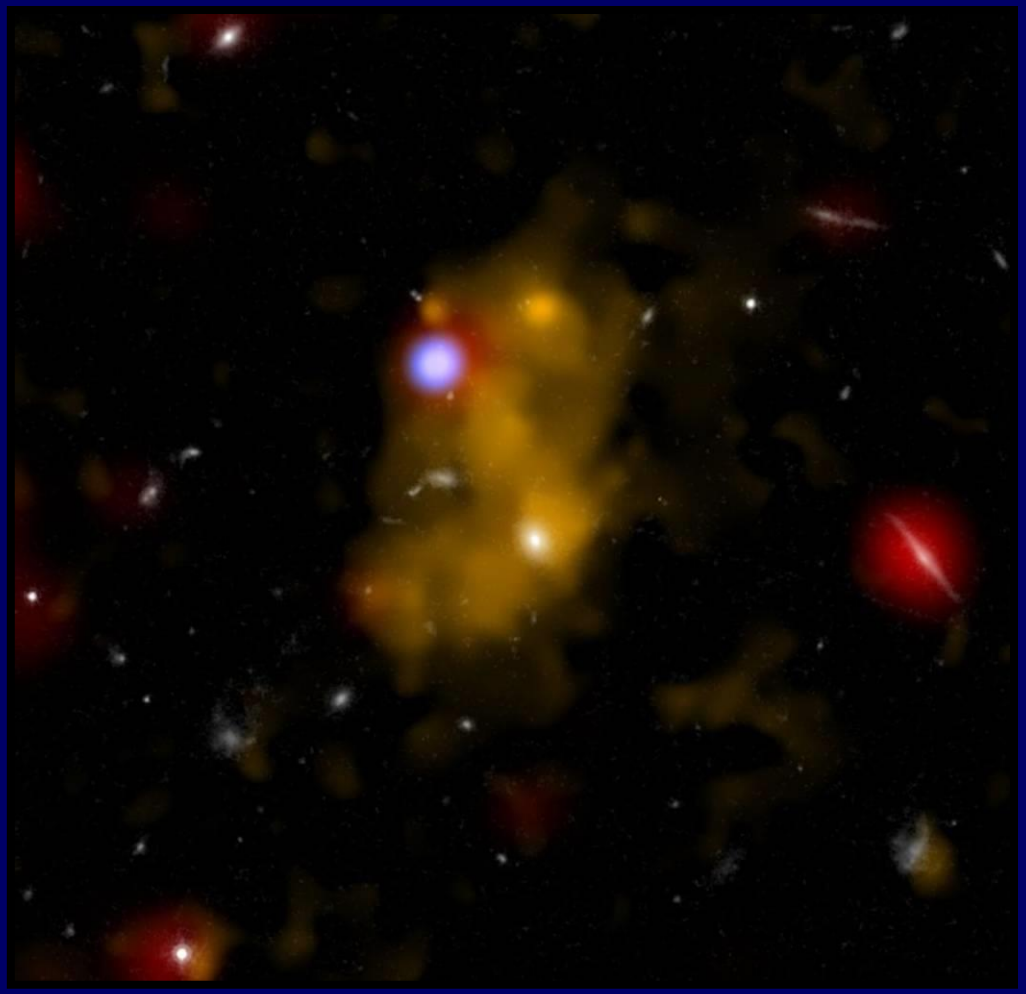
The Crab nebula: remnants of a supernova that was observed in 1054; at its heart is a neutron star

Summary: The origin of elements in the periodic table



Back to our story: Towards the formation of the first galaxies

- The intense ultraviolet radiation emitted by the super-massive Population III stars re-ionized the surrounding medium (stripped electrons from hydrogen atoms)
- Shock waves from the supernovae that marked the end of these stars compressed the surrounding ionized gas triggering a burst of star formation
- Black holes formed from core collapse
- By 250 myr: black holes of at least 400 solar masses were in place
- Ongoing gravitational collapse around these black holes led to the formation of the first galaxies by 400 myr



SSA 22, a region over three times the size of the Milky Way galaxy that contains two large galaxies undergoing vigorous star formation and several satellite galaxies, all shrouded within a cloud of hydrogen gas

Galaxy merger



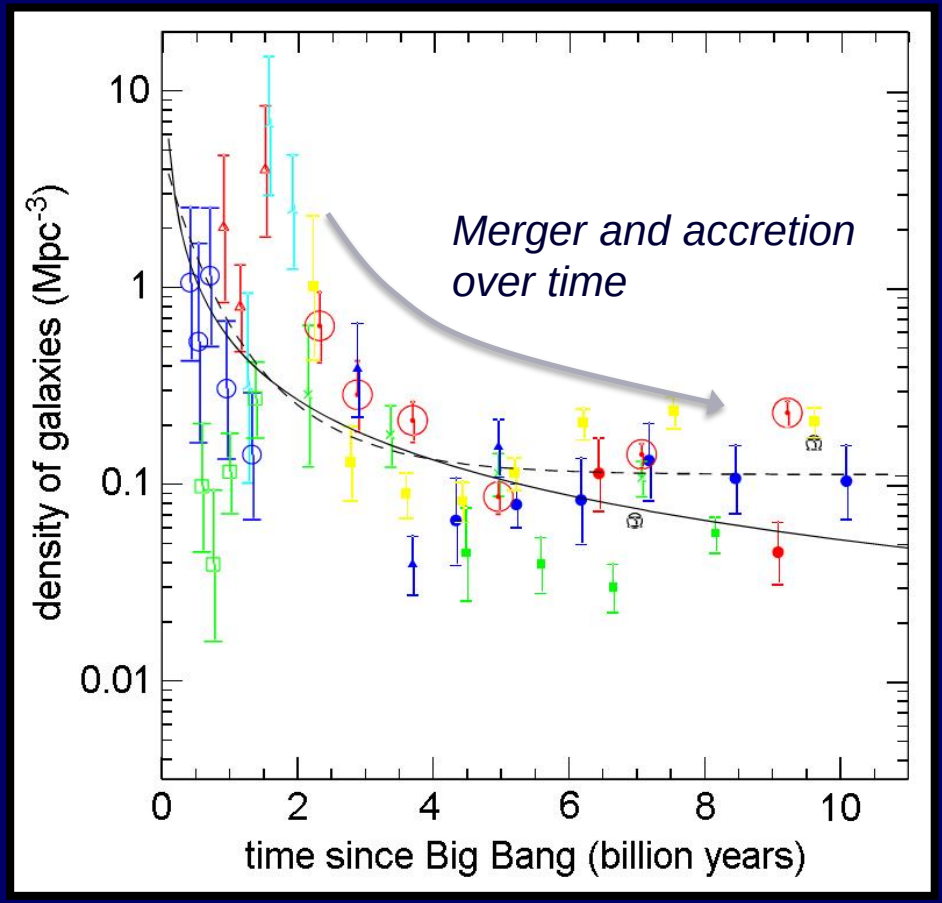
- Once galaxies began to form they collided and merged to form much larger systems.
- Almost every large galaxy has a supermassive black hole at the galaxy's centre (supermassive = 10 billion solar masses).

Galaxy formation



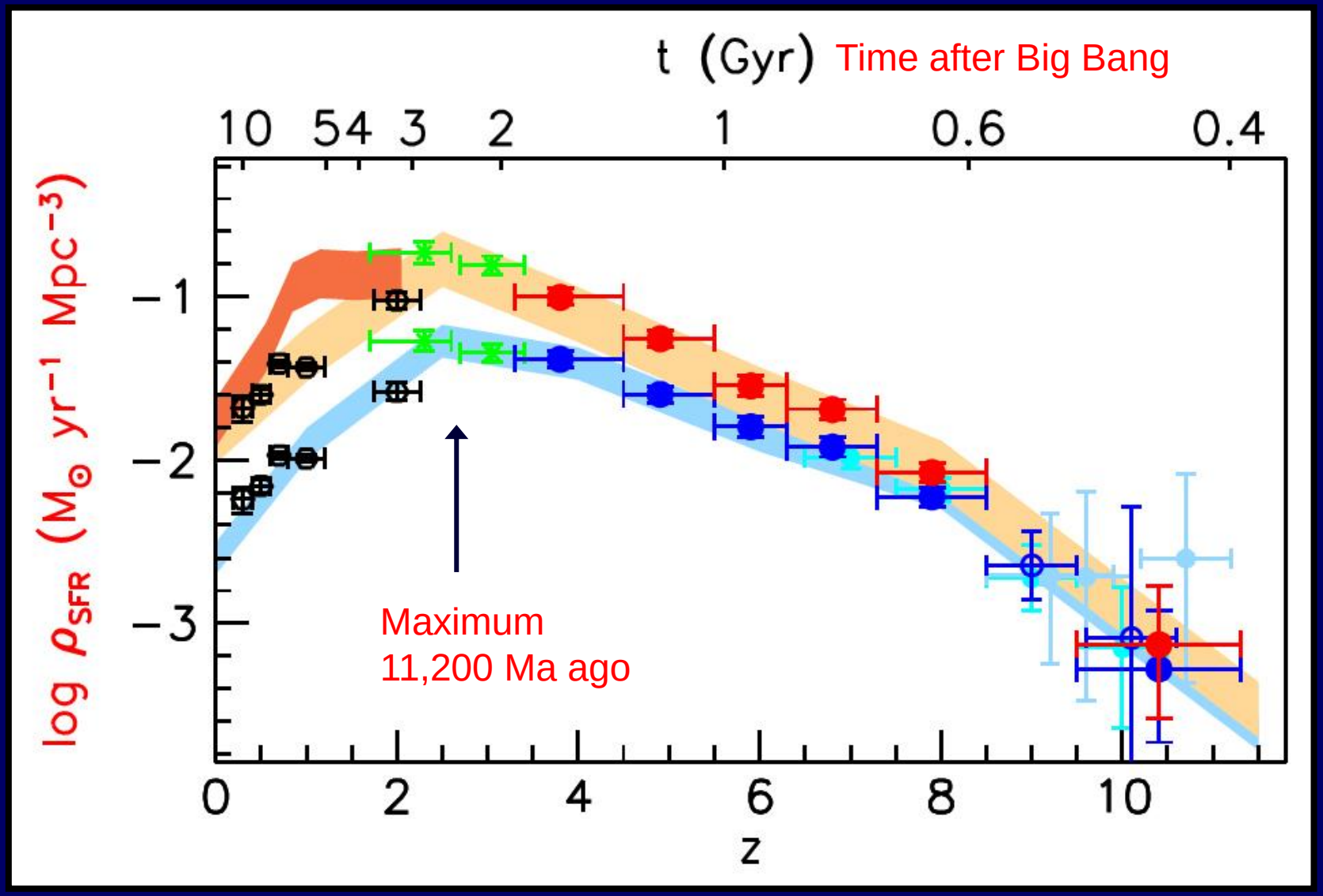
Hubble Ultra-Deep Field image, ~800 million years after the Big Bang (10,000 galaxies)

A galaxy is a gravitationally bound system of stars, stellar remnants, interstellar gas, dust, and dark matter (~100 million stars)



Number of galaxies with time (at least 100 billion in the Universe)

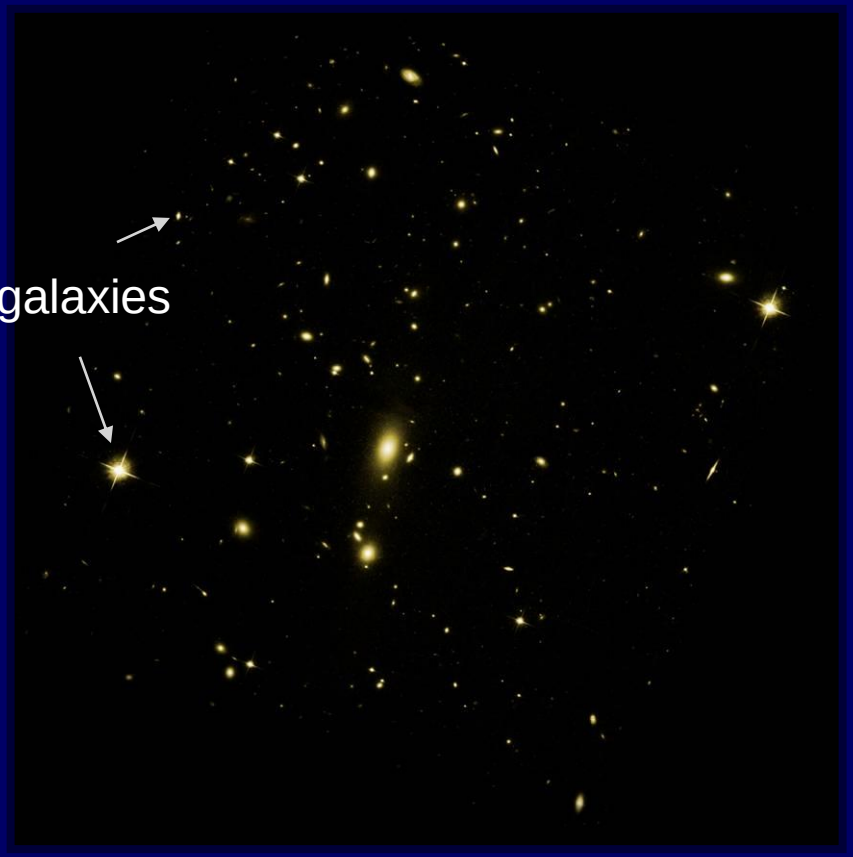
Galaxy merger



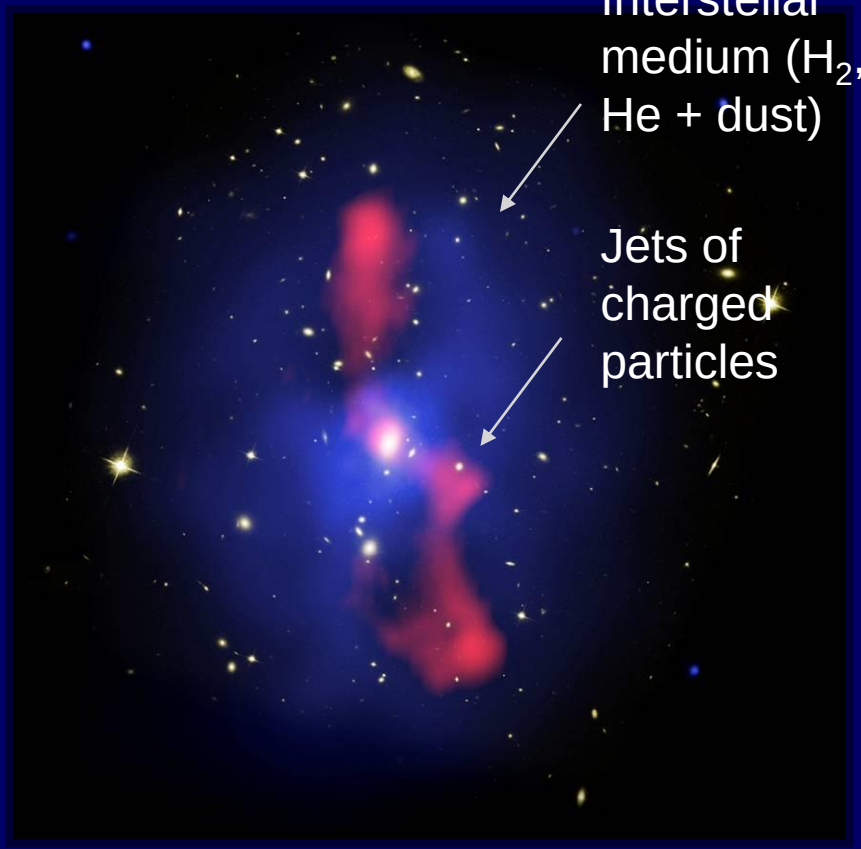
Star formation rate (SFR) with time

Development of galaxy clusters

Galaxy cluster MS 0735.6+7421



Visible light image
(Hubble Space Telescope)



Composite image
(Hubble + NASA's Chandra X-ray observatory + Very Large Array Telescope, New Mexico)

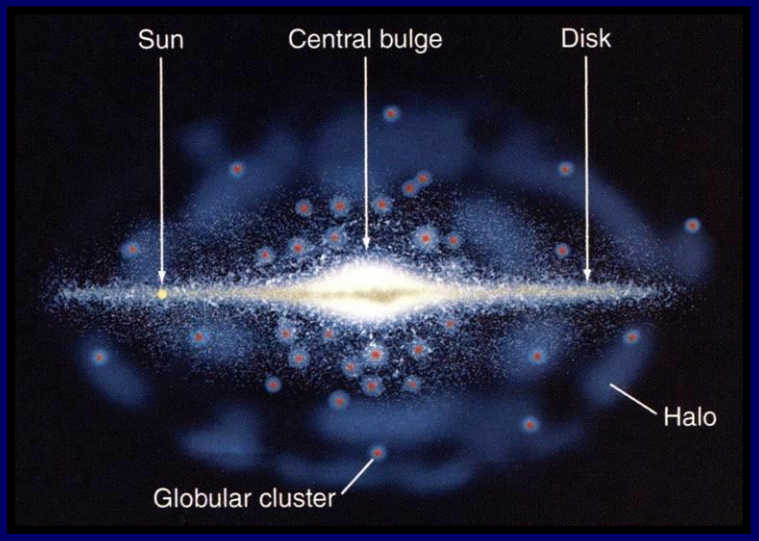
Formation and Evolution of the Milky Way galaxy (13200 Ma)

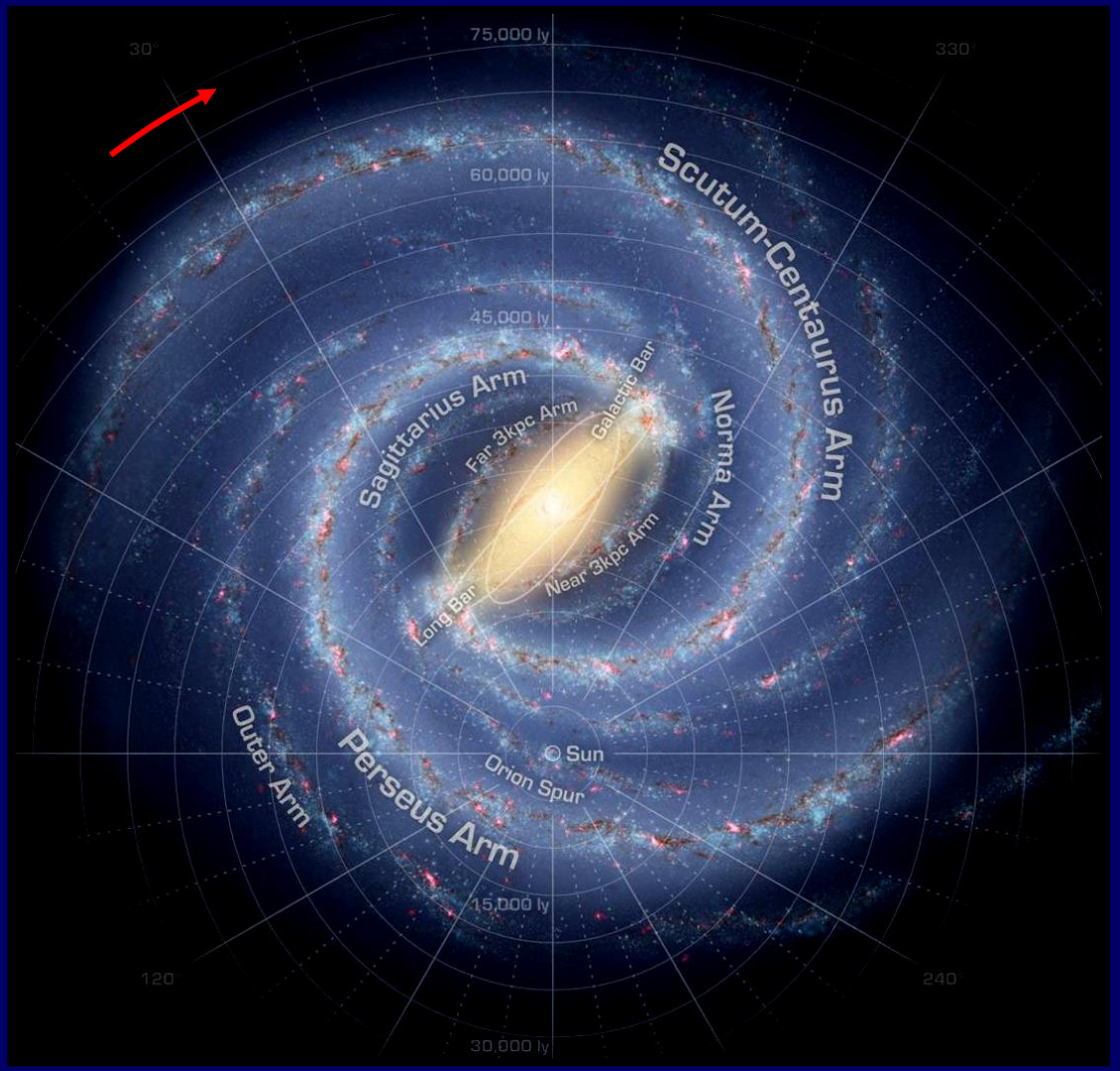


- 100-400 billion stars
- 100-400 billion exoplanets
- 300 million Earth-like, potentially habitable

Milky Way galaxy- geography and geometry

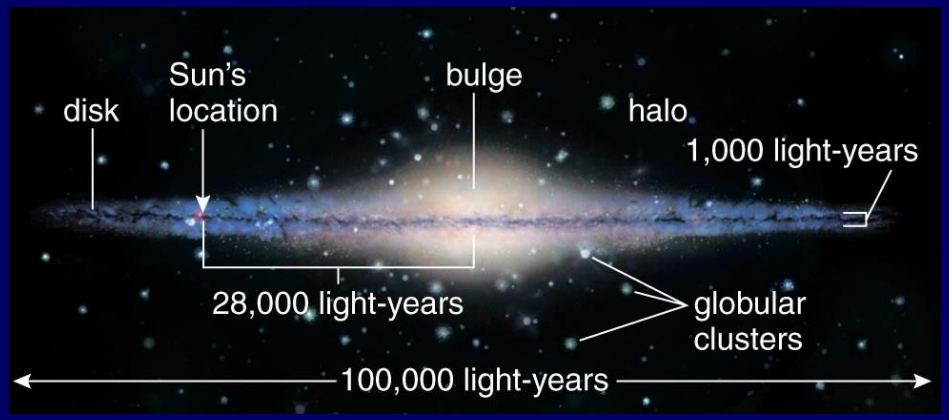
- a rotating, barred spiral galaxy
- disc of stars ~ 100000 light years across by ~ 1000 light years thick
- bulges near the centre
- surrounded by spherical halo containing globular star clusters
- It has a supermassive black hole (Sagittarius A*) at its centre





Artist's impression looking down on the galactic disc (above), and a section through the plane of the disc (left)

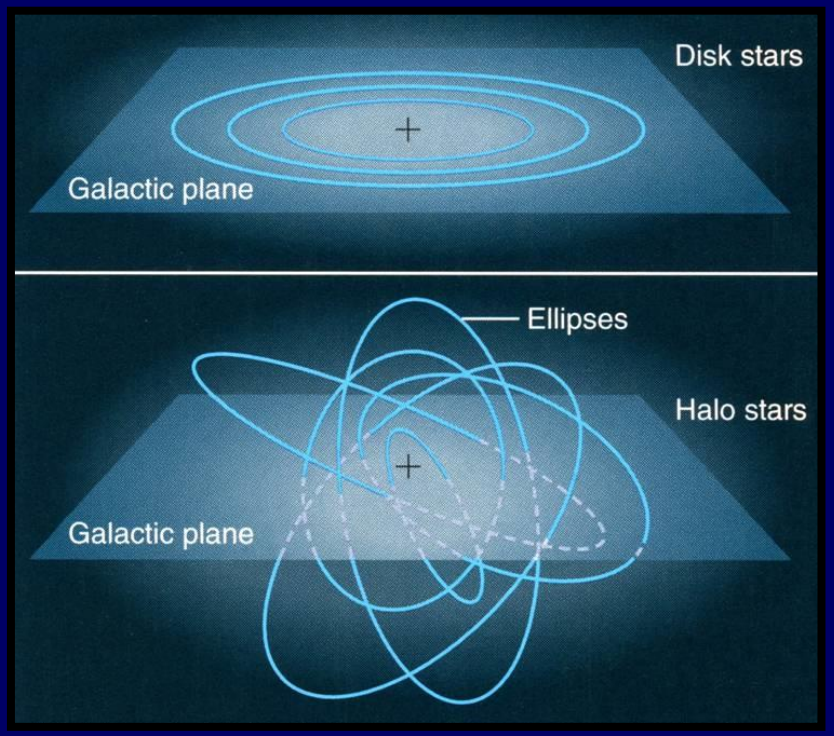
MW Stellar populations- age of stars and orbits



- Young stars are in the disc
- Older stars are distributed within the disc, bulge and halo
- Very old stars are in the globular clusters of the halo

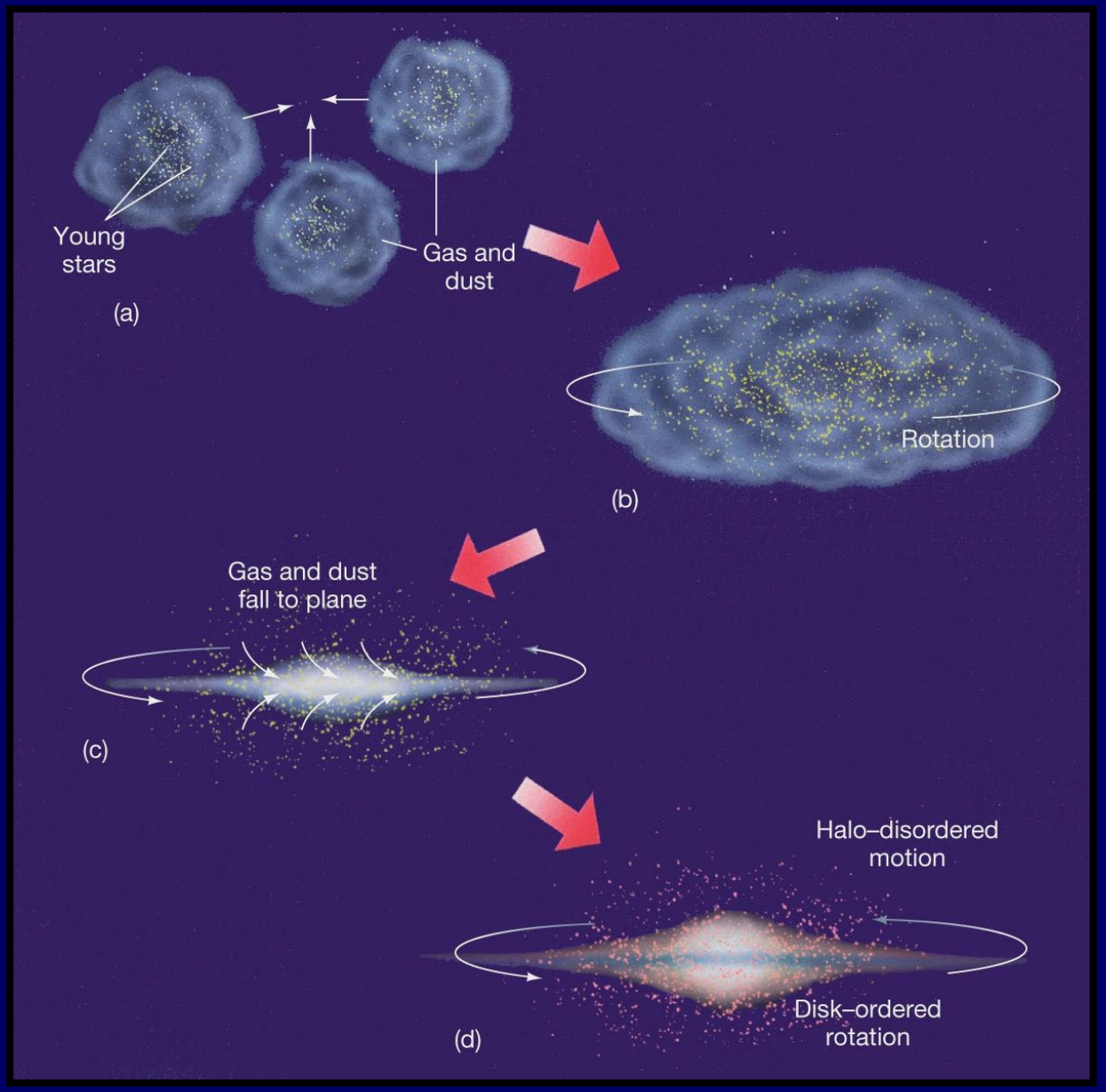
The galactic disc contains most of the galaxy's 200-400 billion stars and nearly all its gas and dust

- Disc stars have approximately circular orbits within the mid-plane of the disc
- Bulge and halo stars have elliptical orbits that are randomly oriented to the galactic plane

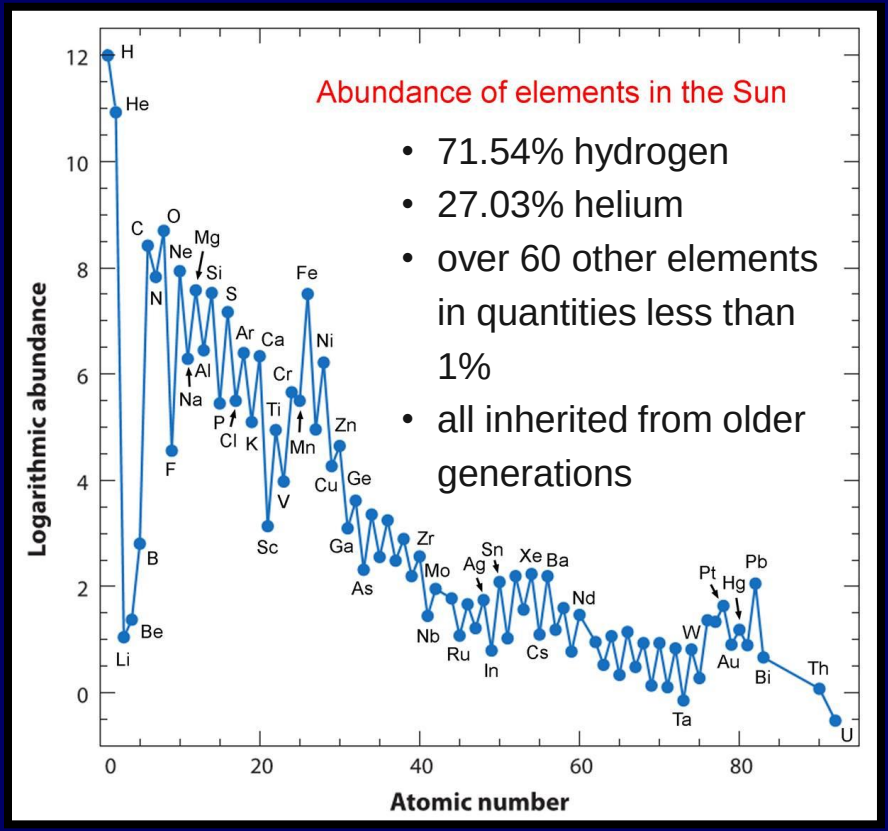


Formation and evolution of the Milky Way galaxy

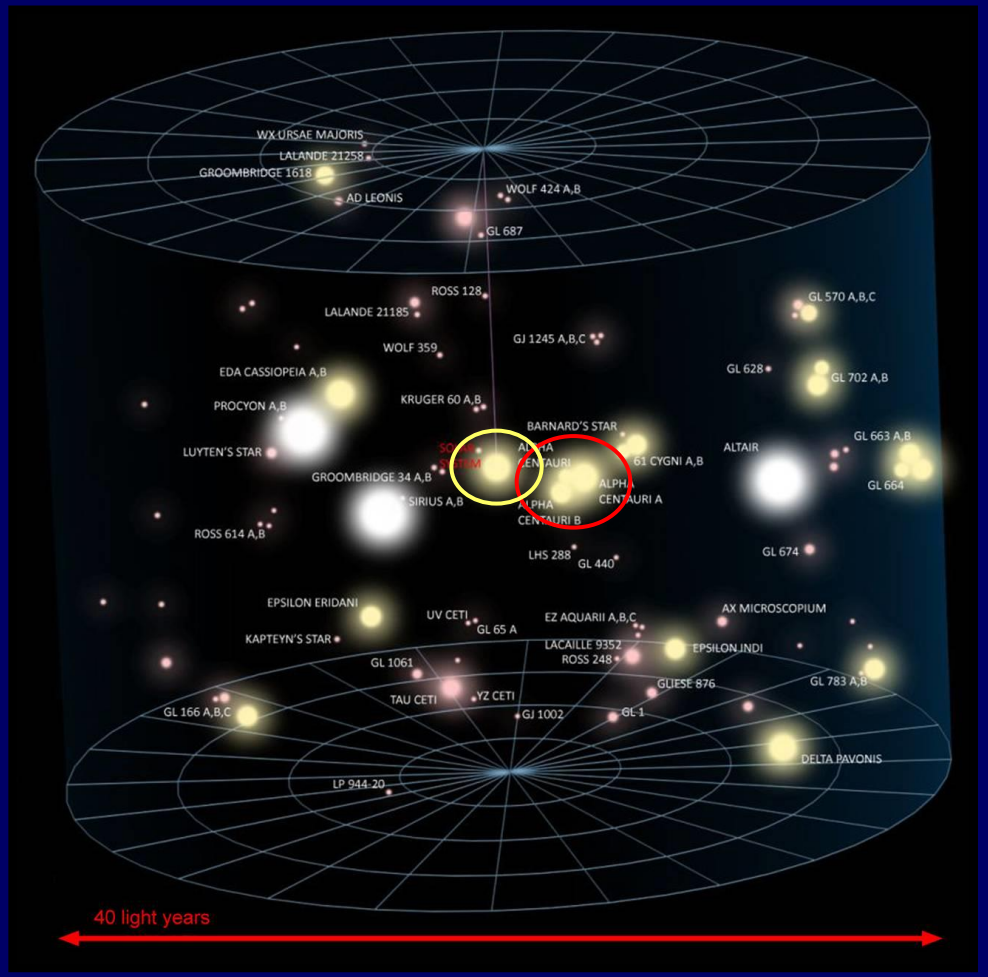
- The Milky Way galaxy began as several over-densities in mass distribution; globular clusters were seeded within these (~13200 Ma)
- Gravitational attractions led to the merger of these over-densities and to the initiation of spinning; stars in the inner halo were forming by ~11500 Ma
- Conservation of angular momentum led to collapse of the gaseous interstellar medium from a spheroid shape to a disc (by 8800 ± 1700 Ma)



Neighbourhood of the Sun within the Milky Way



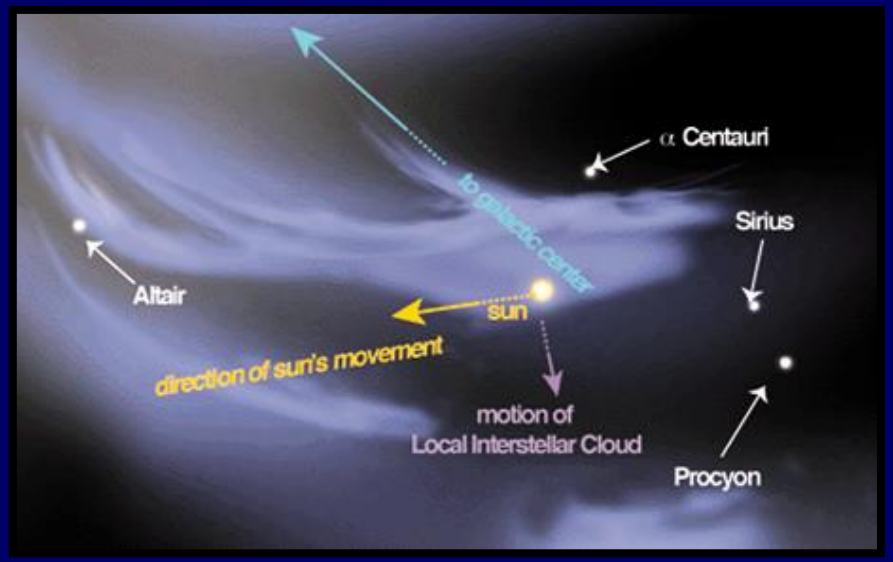
- The Sun is a Population I star formed ~4600 Ma
- Located in the galactic disc ~27000 light years from the galactic centre, just above the galactic mid-plane

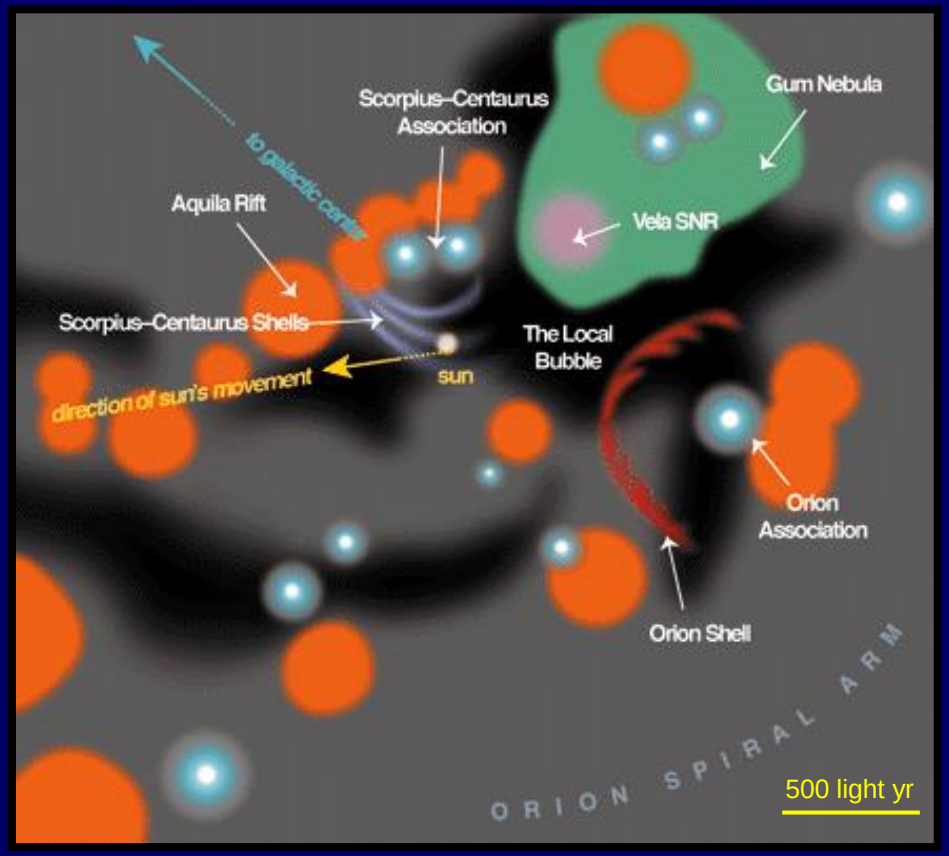


Map of the stars within 40 light years of the Sun; the nearest star is Proxima Centauri, 4.25 light years away

Changing interstellar environment as the Sun orbits

- The Sun is currently in the Local Bubble, a very low density region of the interstellar medium that was sculpted by supernovae
- It is passing through the Local Interstellar Cloud, a slightly higher density region that comprises material flowing away from the sites of several supernovae that have occurred within the Scorpius-Centaurus Star Association over the past 15 myr





- Material from these supernovae, dated at 3.2-1.5 Ma and 8.7-6.5 Ma has been found in deep sea sediments and in lunar soils

Intended Learning Outcome Revision

Now we have completed the class, you should be able to:

- describe how the first stars formed and how these they differ to later stars
- compare and contrast the life-cycle of Sun-sized stars with massive stars
- explain where elements heavier than hydrogen and helium came from and their relationship to processes associated with sun-sized and more massive stars
- describe how galaxies evolve, and in particular our own Milky Way
- be able to place our Sun in the context of the Milky Way and its local interstellar environment